

MIT 18.05

Probability & Statistics

Intuition, Theory, Proofs, Applications, and Exercises

A self-contained companion built around the complete topic arc of MIT 18.05: probability, Bayesian inference, frequentist inference, confidence intervals, bootstrap methods, and regression. Every major idea is developed from a mental picture to formal mathematics, with derivations, real-world modeling examples, and practice.

How to use this book

This guide is designed to be read in layers. First read the **Intuition** box and try to explain the idea in your own words. Then read the formal statement and derivation. Work the example without looking at the solution, and finish with the exercises. If you are revising for an exam, use the checkpoint boxes and the formula atlas at the end.

The emphasis is not on memorizing formulas. A formula is useful only when you know what random experiment it describes, what assumptions make it valid, and how the answer should change when the situation changes. Throughout the book, probability is treated as a language for uncertainty and statistics as the inverse problem: learning about an uncertain mechanism from data produced by it.

Coverage map

The guide follows the complete course arc and includes the computational themes used throughout the class.

Arc	Topics covered
Probability	Sets and counting; probability axioms; conditional probability; independence; Bayes' theorem; random variables; Bernoulli, binomial, geometric, Poisson, uniform, exponential and normal models; expectation; variance; quantiles; histograms; transformations; joint distributions; covariance; correlation; law of large numbers; central limit theorem.
Bayesian inference	Likelihood and MLE; discrete and continuous priors; posterior distributions; posterior odds; predictive probabilities; Beta and Normal conjugate priors; prior choice; credible/probability intervals.
Frequentist inference	Sampling distributions; NHST; rejection regions; p-values; Type I/II error; power; z-tests; one- and two-sample t-tests; chi-square tests; ANOVA; comparison with Bayesian inference.
Intervals and regression	Confidence intervals under normal and non-normal sampling; three interpretations of intervals; empirical and parametric bootstrap; simple and multiple linear regression; least squares; uncertainty and diagnostics.
Computation	Simulation, visualization, and representative R workflows for probability, inference, resampling, and regression.

Contents

How to use this book	ii
I Probability: Building a Language for Uncertainty	1
1 Counting, Sets, and Sample Spaces	2
1.1 Sets as events	2
1.2 The multiplication rule	2
1.3 Permutations and combinations	2
1.4 Inclusion-exclusion and derangements	3
2 Probability Axioms, Rules, and Odds	4
2.1 Kolmogorov axioms	4
2.2 Equally likely finite outcomes	4
2.3 Odds	4
2.4 Simulation as a probability microscope	5
3 Conditional Probability, Independence, and Bayes' Theorem	6
3.1 Conditional probability	6
3.2 Independence	6
3.3 Law of total probability and Bayes	6
3.4 Likelihood ratios and odds form	7
4 Discrete Random Variables and Distribution Functions	8
4.1 PMF and CDF	8
4.2 Functions of a random variable	8
4.3 Quantiles	9
5 The Core Discrete Distributions	10
5.1 Bernoulli and binomial	10
5.2 Geometric	10
5.3 Poisson	10

6	Expectation: The Center of a Random Quantity	12
6.1	Definition and linearity	12
6.2	LOTUS	12
6.3	Indicator variables	13
6.4	Means of standard distributions	13
7	Variance, Standard Deviation, and Moments	14
7.1	Definition and computational form	14
7.2	Independent sums	14
7.3	Standard results	15
8	Continuous Random Variables: PDFs, CDFs, and Probability as Area	16
8.1	Density and distribution function	16
8.2	Density can exceed one	16
8.3	Transformation by a monotone function	17
9	Uniform, Exponential, and Normal Distributions	18
9.1	Uniform distribution	18
9.2	Exponential distribution	18
9.3	Normal distribution	18
10	Continuous Expectation, Variance, Quantiles, and Histograms	20
10.1	Expectation and LOTUS	20
10.2	Tail-integral formula	20
10.3	Quantiles and robust summaries	21
10.4	Histograms	21
11	Manipulating and Combining Random Variables	22
11.1	Linear transformations	22
11.2	Sums and convolution	22
11.3	Maxima and minima	22
12	Law of Large Numbers and Central Limit Theorem	24
12.1	Weak law of large numbers	24
12.2	Central limit theorem	24
12.3	Normal approximation to binomial	25
13	Joint Distributions, Marginals, and Conditional Distributions	26
13.1	Discrete joint PMF	26
13.2	Continuous joint density	26
13.3	Conditional expectation	27

14 Covariance, Correlation, and Dependence	28
14.1 Definitions	28
14.2 Variance of a sum	28
14.3 Independence implies zero covariance	28
14.4 Why correlation is between -1 and 1	29
II Statistics I: Learning from Data with Bayesian Inference	30
15 From Probability to Statistics: Data, Models, Likelihood, and MLE	31
15.1 Data, parameter, statistic	31
15.2 Probability versus likelihood	31
15.3 Maximum likelihood estimation	31
15.4 MLEs for familiar models	32
15.5 Exploratory plots	32
16 Bayesian Updating with Discrete Priors	33
16.1 Discrete parameter space	33
16.2 Sequential updating	33
16.3 Posterior summaries	34
17 Bayesian Prediction, Posterior Odds, and Evidence	35
17.1 Posterior predictive distribution	35
17.2 Odds and Bayes factors	35
17.3 Prediction versus plug-in estimation	36
18 Bayesian Updating with Continuous Priors	37
18.1 Bayes' rule for densities	37
18.2 Kernel recognition	37
18.3 Continuous data with continuous priors	38
19 Conjugate Priors: Beta-Binomial and Normal-Normal	39
19.1 Beta distribution	39
19.2 Beta-Binomial updating	39
19.3 Pseudo-count interpretation	40
19.4 Normal-normal precision weighting	40
20 Choosing Priors and Sensitivity Analysis	41
20.1 Informative, weakly informative, and reference-like priors	41
20.2 Prior predictive checking	41
20.3 Sensitivity	41
21 Bayesian Probability Intervals and Credible Intervals	43

21.1	Equal-tail intervals	43
21.2	Highest posterior density intervals	43
21.3	Decision-focused probabilities	43
III	Statistics II: Frequentist Inference and Hypothesis Testing	45
22	The Frequentist Framework and Sampling Distributions	46
22.1	Estimator and sampling distribution	46
22.2	Standard error	46
22.3	Unbiased sample variance	46
23	Null Hypothesis Significance Testing, p-Values, and z-Tests	48
23.1	Ingredients of a test	48
23.2	p-values	48
23.3	One-sample z-test	49
23.4	Rejection regions and p-values are equivalent views	49
23.5	Power and sample size	49
24	t-Distributions and One- and Two-Sample t-Tests	50
24.1	Student's t distribution	50
24.2	One-sample t-test	50
24.3	Paired t-test	50
24.4	Two independent samples	51
25	Chi-Square Distributions and Chi-Square Tests	52
25.1	Chi-square distribution	52
25.2	Goodness-of-fit test	52
25.3	Test of independence in a contingency table	52
25.4	Connection to variance inference	53
26	One-Way ANOVA and the F Distribution	54
26.1	Model	54
26.2	Variance decomposition	54
26.3	F statistic	54
27	Bayesian and Frequentist Inference: Two Logics	56
27.1	What is random?	56
27.2	Neither logic removes modeling assumptions	56
27.3	Why answers can resemble each other	56

IV Statistics III: Confidence Intervals, Bootstrap, and Regression	58
28 Confidence Intervals: Construction and Interpretation	59
28.1 Pivot idea	59
28.2 Unknown variance: t interval	59
28.3 Large-sample intervals for non-normal means	60
28.4 Proportion interval	60
28.5 Three views of confidence intervals	60
29 Bootstrap Confidence Intervals	61
29.1 Empirical distribution	61
29.2 The bootstrap principle	61
29.3 Bootstrap standard error	61
29.4 Percentile and basic intervals	62
29.5 Parametric bootstrap	62
30 Simple Linear Regression	63
30.1 Model	63
30.2 Least-squares estimates	63
30.3 Slope and correlation	64
30.4 Residuals and R^2	64
30.5 Inference for slope	64
31 Multiple Regression, Least Squares Geometry, and Diagnostics	65
31.1 Matrix model	65
31.2 Interpreting coefficients	65
31.3 Diagnostics	66
31.4 Extrapolation and model scope	66
32 Simulation, Visualization, and R Workflows	67
32.1 The d/p/q/r pattern	67
32.2 Histograms and scatter plots	67
32.3 Bayesian grid update	68
32.4 Classical tests and intervals	68
32.5 Monte Carlo error	68
V Deep Connections and Exam-Level Synthesis	69
33 A Unifying View: Four Distributions You Must Not Confuse	70
33.1 Population distribution	70
33.2 Likelihood	70

33.3 Sampling distribution	70
33.4 Posterior distribution	70
34 Modeling Checklist: From Story to Mathematics	71
35 Comprehensive Mixed Practice Set	72
 VI Reference Appendices	 75
A Proof Toolbox	76
A.1 Linearity of expectation	76
A.2 Markov and Chebyshev inequalities	76
A.3 Jensen's inequality	76
A.4 Cauchy-Schwarz	76
A.5 Law of total expectation and variance	77
 B Distribution Atlas	 78
 C Formula Atlas	 79
 D Conceptual Traps and How to Avoid Them	 80
 E Study Roadmap	 81
 Course Coverage Reference	 82

Part I

Probability: Building a Language for Uncertainty

Counting, Sets, and Sample Spaces

Intuition

Before we assign probabilities, we need to know *what can happen*. Counting is the bookkeeping layer of probability. If a password has four independent digits, there are 10^4 possible strings because each position multiplies the number of possibilities. If order does not matter, we must divide away the repeated orderings. This simple idea becomes permutations, combinations, the binomial coefficient, and much of elementary probability.

1.1 Sets as events

A sample space Ω is the set of all possible outcomes. An event $A \subseteq \Omega$ is a statement that may or may not occur. The operations $A \cup B$, $A \cap B$, and A^c translate “ A or B ”, “ A and B ”, and “not A ”. De Morgan’s laws,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c,$$

are not just symbolic tricks: they let us replace a complicated condition with an easier complementary one.

1.2 The multiplication rule

If a procedure has n_1 choices for step 1, n_2 choices for step 2 after every step-1 choice, and so on, then the total number of outcomes is $n_1 n_2 \cdots n_k$.

Proof / derivation

Represent every complete outcome by a path through a decision tree. Level 1 has n_1 branches. Every branch then creates n_2 children, so there are $n_1 n_2$ partial paths after two levels. Continuing inductively multiplies the path count by n_j at level j .

1.3 Permutations and combinations

The number of ordered selections of k distinct objects from n is

$$P(n, k) = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

If order does not matter, each chosen set of k objects was counted $k!$ times, once per ordering. Therefore

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

With repeated categories, a string of length n containing counts $n_1, \dots, n_r$ has multinomial count

$$\frac{n!}{n_1! \cdots n_r!}.$$

Worked example

A committee of 4 is chosen from 7 engineers and 5 designers, with at least 2 designers. Count by cases:

$$\binom{5}{2} \binom{7}{2} + \binom{5}{3} \binom{7}{1} + \binom{5}{4} \binom{7}{0} = 210 + 70 + 5 = 285.$$

The lesson is modeling: identify whether order matters and whether constraints are easier handled by cases or complements.

1.4 Inclusion-exclusion and derangements

For finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets, add individual sizes, subtract pairwise intersections, then add back the triple intersection. The alternating pattern continues. A classic application is the number of derangements D_n , permutations with no fixed point:

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \approx \frac{n!}{e}.$$

This comes directly from inclusion-exclusion applied to the events “person i gets their own item.”

Exercises

1. How many 8-character strings over 26 letters contain exactly three A's?
2. A class has 12 students. How many ways can we split them into three *unlabeled* groups of four?
3. A random permutation of 6 objects is formed. Compute the probability that nobody remains in the original position.
4. Explain why $\sum_{k=0}^n \binom{n}{k} = 2^n$ using a counting argument rather than algebra.

Selected solutions

1. Choose the three positions for A, then fill the rest with 25 non-A letters: $\binom{8}{3} 25^5$.
2. First form labeled groups: $\binom{12}{4} \binom{8}{4} \binom{4}{4}$, then divide by $3!$: $12!/(4!^3 3!)$.
3. $D_6/6! = 265/720 \approx 0.3681$.
4. Both sides count subsets of an n -element set: the left counts them by size, the right gives each element two choices, in or out.

Probability Axioms, Rules, and Odds

Intuition

Probability is a consistent way to distribute one unit of belief or long-run frequency over possible outcomes. The three axioms are minimal, but they force almost every familiar rule. They also reveal why “equally likely outcomes” is a special model, not the definition of probability.

2.1 Kolmogorov axioms

For every event A in a sample space Ω ,

$$\mathbb{P}(A) \geq 0, \quad \mathbb{P}(\Omega) = 1,$$

and for pairwise disjoint events $A_1, A_2, \dots$,

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

From these axioms we derive

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof / derivation

Because $\Omega = A \cup A^c$ is a disjoint union, $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$, proving the complement rule. For the union rule, decompose $B = (B \setminus A) \cup (A \cap B)$. Then $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$ and $A \cup B = A \cup (B \setminus A)$ disjointly.

2.2 Equally likely finite outcomes

If a finite experiment has N outcomes, all with probability $1/N$, and A contains m of them, then $\mathbb{P}(A) = m/N$. This is why counting and probability fit together so naturally for shuffled cards, fair dice, and uniformly random permutations.

2.3 Odds

Odds in favor of A are

$$O(A) = \frac{\mathbb{P}(A)}{1 - \mathbb{P}(A)}.$$

If odds are $a : b$, then $\mathbb{P}(A) = a/(a + b)$. Odds are useful in Bayesian updating because independent pieces of evidence multiply odds by likelihood ratios.

Worked example

Two fair dice are rolled. Let A be “sum at least 10”. There are 36 equally likely ordered pairs and 6 favorable pairs: $(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)$. Hence $\mathbb{P}(A) = 1/6$. The odds in favor are $(1/6) : (5/6) = 1 : 5$.

Common pitfall

Do not count outcomes as if they were equally likely unless your model says they are. The sums of two dice are *not* equally likely: sum 7 has six representations while sum 2 has one.

2.4 Simulation as a probability microscope

Simulation does not replace mathematics; it lets us test models, approximate difficult probabilities, and build intuition. If I_j is 1 when trial j succeeds, then the simulated frequency $\hat{p} = (I_1 + \cdots + I_n)/n$ approaches p as n grows. Later the law of large numbers will explain why.

Exercises

1. In a 52-card deck, find the probability of at least one ace in a 5-card hand using a complement.
2. Events A and B satisfy $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B) = 0.4$, $\mathbb{P}(A \cup B) = 0.7$. Find $\mathbb{P}(A \cap B)$.
3. Convert probability 0.8 to odds and odds 3 : 7 to probability.
4. Give an example where counting favorable outcomes over total outcomes is invalid because outcomes are not equally likely.

Selected solutions

1. $1 - \binom{48}{5}/\binom{52}{5}$.
2. $0.5 + 0.4 - 0.7 = 0.2$.
3. $0.8/(0.2) = 4 : 1$; $3/(3 + 7) = 0.3$.
4. Example: selecting a sum from two dice by rolling the dice; the possible sums 2 through 12 do not have equal probability.

Conditional Probability, Independence, and Bayes' Theorem

Intuition

Conditioning means changing the universe after learning information. If we learn B occurred, outcomes outside B are erased, and probabilities are renormalized inside B . Bayes' theorem simply asks the reverse question: if we know how likely evidence is under each cause, what should we believe about the cause after seeing the evidence?

3.1 Conditional probability

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Rearranging gives the multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

For a sequence,

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \prod_{j=2}^n \mathbb{P}(A_j \mid A_1 \cap \cdots \cap A_{j-1}).$$

3.2 Independence

A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If both probabilities are positive, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. Independence is a model statement: learning B carries no information about A . It is much stronger than the events merely being different.

3.3 Law of total probability and Bayes

If $H_1, \dots, H_k$ partition the sample space, then

$$\mathbb{P}(E) = \sum_{i=1}^k \mathbb{P}(E \mid H_i)\mathbb{P}(H_i).$$

Therefore

$$\mathbb{P}(H_j | E) = \frac{\mathbb{P}(E | H_j)\mathbb{P}(H_j)}{\sum_i \mathbb{P}(E | H_i)\mathbb{P}(H_i)}.$$

Proof / derivation

Because the H_i partition Ω , the event E is the disjoint union $\bigcup_i (E \cap H_i)$. Add probabilities and use $\mathbb{P}(E \cap H_i) = \mathbb{P}(E | H_i)\mathbb{P}(H_i)$. Bayes' theorem then follows by applying the definition $\mathbb{P}(H_j | E) = \mathbb{P}(H_j \cap E)/\mathbb{P}(E)$.

Worked example

Suppose 1% of a population has a condition. A test has sensitivity 95% and false-positive rate 5%. For a positive test,

$$\mathbb{P}(D | +) = \frac{0.95(0.01)}{0.95(0.01) + 0.05(0.99)} \approx 0.161.$$

A positive result is meaningful, but it does not imply 95% probability of disease. The low base rate matters. This is the base-rate effect.

3.4 Likelihood ratios and odds form

Bayes becomes especially transparent in odds:

$$\underbrace{\frac{\mathbb{P}(H | E)}{\mathbb{P}(H^c | E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}}_{\text{prior odds}} \times \underbrace{\frac{\mathbb{P}(E | H)}{\mathbb{P}(E | H^c)}}_{\text{likelihood ratio}}.$$

Evidence multiplies odds. Multiple conditionally independent observations multiply multiple likelihood ratios.

Exercises

- Two cards are drawn without replacement. Find the probability the second is an ace given the first is an ace.
- A spam filter flags 90% of spam and 4% of legitimate mail. If 20% of messages are spam, find $\mathbb{P}(\text{spam} | \text{flag})$.
- Show that if A and B are independent, then A^c and B are independent.
- Explain why pairwise independence does not necessarily imply mutual independence; construct a three-event example using two fair bits.

Selected solutions

- $3/51 = 1/17$.
- $0.9(0.2)/(0.9(0.2) + 0.04(0.8)) \approx 0.8491$.
- $\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(B)[1 - \mathbb{P}(A)] = \mathbb{P}(A^c)\mathbb{P}(B)$.
- Let bits X, Y be independent fair bits and define events $A = \{X = 0\}$, $B = \{Y = 0\}$, $C = \{X \oplus Y = 0\}$. Each pair is independent, but $A \cap B$ forces C .

Discrete Random Variables and Distribution Functions

Intuition

A random variable is not a “random number” floating in space. It is a function that translates an outcome into a numerical feature we care about: number of failures, revenue, queue length, or time slots used. Once we define that function, its distribution summarizes how uncertainty in the outcome becomes uncertainty in the number.

4.1 PMF and CDF

A discrete random variable X has probability mass function

$$p_X(x) = \mathbb{P}(X = x), \quad \sum_x p_X(x) = 1.$$

Its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

The CDF is defined for every real x , is nondecreasing, right-continuous, and approaches 0 and 1 at the two tails. For discrete variables it is a step function.

Formal statement

A distribution is the complete probability law of a random variable. Two completely different physical experiments can generate the same distribution. Statistical modeling often works at this distributional level.

4.2 Functions of a random variable

If $Y = g(X)$, then Y inherits probabilities from X :

$$\mathbb{P}(Y = y) = \sum_{x: g(x)=y} \mathbb{P}(X = x).$$

This “push forward” idea is fundamental. We will use continuous analogues later.

Worked example

Roll two fair dice and let X be the sum. Then

$$p_X(2), \dots, p_X(12) = \frac{1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1}{36}.$$

The triangular shape comes from counting how many ordered pairs map to each sum.

4.3 Quantiles

A q -quantile is a threshold x_q such that roughly a fraction q of the distribution lies at or below it. For discontinuous CDFs, quantiles need a careful convention; a standard choice is

$$x_q = \inf\{x : F_X(x) \geq q\}.$$

Medians are 0.5-quantiles. Quantiles are often more robust than means because they depend on order rather than the magnitude of extreme values.

Exercises

1. A fair die is rolled. Let X be the square of the result. Write the PMF and CDF.
2. Let X have PMF $p(0) = 0.2, p(1) = 0.5, p(2) = 0.3$. Find the distribution of $Y = (X - 1)^2$.
3. Construct two physically different experiments whose random variables both have a Bernoulli(0.3) distribution.
4. For the PMF in problem 2, find the 0.25-, 0.5-, and 0.9-quantiles under the infimum convention.

Selected solutions

1. $X \in \{1, 4, 9, 16, 25, 36\}$ with mass $1/6$ each.
2. $Y = 1$ when $X \in \{0, 2\}$, so $\mathbb{P}(Y = 1) = 0.5$ and $\mathbb{P}(Y = 0) = 0.5$.
3. Example: success of a biased coin; whether a randomly sampled component is defective when the defect rate is 0.3.
4. CDF jumps to 0.2 at 0, 0.7 at 1, 1 at 2, hence quantiles are 1, 1, 2.

The Core Discrete Distributions

Intuition

Distributions are reusable stories. Bernoulli describes one yes/no trial. Binomial counts successes in a fixed number of independent trials. Geometric counts how long we wait for the first success. Poisson counts rare, roughly independent events in a fixed exposure window. Recognizing the story is more important than memorizing the formula.

5.1 Bernoulli and binomial

If $X \sim \text{Bern}(p)$, then $\mathbb{P}(X = 1) = p$, $\mathbb{P}(X = 0) = 1 - p$. If X counts successes among n independent Bernoulli(p) trials, then

$$X \sim \text{Bin}(n, p), \quad \mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

The combinatorial factor chooses which k trials succeed; the probability factor assigns the probability of each such pattern.

5.2 Geometric

If X is the trial number of the first success,

$$\mathbb{P}(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, \dots$$

It is memoryless:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Proof / derivation

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > s + t) / \mathbb{P}(X > s) = (1 - p)^{s+t} / (1 - p)^s = (1 - p)^t.$$

5.3 Poisson

If $X \sim \text{Pois}(\lambda)$,

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

The parameter λ is both the mean count and the variance. A common origin is a binomial rare-event limit: if $n \rightarrow \infty$, $p \rightarrow 0$, and $np \rightarrow \lambda$, then $\text{Bin}(n, p)$ approaches $\text{Pois}(\lambda)$.

Poisson limit sketch

Take $p = \lambda/n$. For fixed k ,

$$\binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

Now $\binom{n}{k}/n^k \rightarrow 1/k!$, $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$, and the remaining factor tends to 1. Multiplying gives $e^{-\lambda}\lambda^k/k!$.

Worked example

A server receives an average of 12 independent requests per minute. A Poisson model gives the probability of no request in 10 seconds. The mean for 10 seconds is $\lambda = 12/6 = 2$, so $\mathbb{P}(X = 0) = e^{-2} \approx 0.1353$.

Exercises

1. A fair coin is flipped 20 times. Find the probability of exactly 12 heads.
2. A component succeeds independently with probability 0.02 each test. Find the probability the first success occurs after 30 tests.
3. Calls arrive at 3 per hour under a Poisson model. Find the probability of at least 2 calls in 30 minutes.
4. Explain which assumption fails if a Poisson model is used for customers arriving together in large tour groups.

Selected solutions

1. $\binom{20}{12}2^{-20}$. 2. 0.98^{30} . 3. With $\lambda = 1.5$, $1 - e^{-1.5}(1 + 1.5)$. 4. Arrivals are clustered, so the independent-small-increment picture is violated; variance is often larger than the Poisson mean.

Expectation: The Center of a Random Quantity

Intuition

Expectation is a probability-weighted balance point. It is not necessarily a value the variable can take. If a game pays x dollars in outcome x , the expected value is the long-run average payoff across repeated plays. More deeply, expectation is a linear operator, and that linearity is one of the most powerful simplifiers in probability.

6.1 Definition and linearity

For discrete X ,

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

For any constants a, b ,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

More generally, even when variables are dependent,

$$\mathbb{E}[X_1 + \cdots + X_n] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n].$$

Proof / derivation

For two variables,

$$\mathbb{E}[X + Y] = \sum_{x,y} (x + y) p_{X,Y}(x, y) = \sum_x x \sum_y p_{X,Y}(x, y) + \sum_y y \sum_x p_{X,Y}(x, y),$$

which equals $\mathbb{E}[X] + \mathbb{E}[Y]$ because the inner sums are the marginal probabilities.

6.2 LOTUS

The law of the unconscious statistician says

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$$

for discrete X . We do not need to derive the distribution of $g(X)$ first.

6.3 Indicator variables

For an event A , define $I_A = 1$ if A occurs and 0 otherwise. Then

$$\mathbb{E}[I_A] = \mathbb{P}(A).$$

This turns counting into expectation. If N counts how many of several properties occur, write $N = I_1 + \cdots + I_m$ and use linearity, often without needing independence.

Worked example

In a random permutation of n objects, let I_i indicate that item i stays fixed. Then $\mathbb{P}(I_i = 1) = 1/n$. The number of fixed points is $N = \sum_i I_i$, so

$$\mathbb{E}[N] = \sum_i \frac{1}{n} = 1.$$

No derangement formula is needed. On average, every random permutation has one fixed point, even though the actual number varies.

6.4 Means of standard distributions

$$\mathbb{E}[\text{Bern}(p)] = p, \quad \mathbb{E}[\text{Bin}(n, p)] = np, \quad \mathbb{E}[\text{Geom}(p)] = \frac{1}{p}, \quad \mathbb{E}[\text{Pois}(\lambda)] = \lambda.$$

For the binomial, the cleanest proof writes it as a sum of n Bernoulli indicators.

Exercises

1. A lottery pays 100 with probability 0.01 and loses 2 otherwise. Find the expected profit.
2. Prove $\mathbb{E}[\text{Bin}(n, p)] = np$ using indicators.
3. Ten people independently choose one of 10 seats uniformly. What is the expected number of empty seats?
4. A graph has m edges. Each vertex is independently colored red/blue with equal probability. Find the expected number of edges whose endpoints have different colors.

Selected solutions

1. $100(0.01) - 2(0.99) = -0.98$. 2. If $X = \sum_{i=1}^n I_i$, then $\mathbb{E}X = \sum p = np$. 3. Let I_j indicate seat j is empty. $\mathbb{P}(I_j = 1) = (9/10)^{10}$, so the expectation is $10(9/10)^{10}$. 4. Each edge is bichromatic with probability $1/2$, so expected count is $m/2$.

Variance, Standard Deviation, and Moments

Intuition

The mean says where a distribution balances; variance says how much uncertainty remains around that balance. Squaring deviations prevents cancellation and penalizes large departures. Standard deviation puts the result back in the original units.

7.1 Definition and computational form

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2].$$

Expanding gives the useful identity

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Proof / derivation

Let $\mu = \mathbb{E}X$. Then

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu^2 + \mu^2.$$

For constants a, b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Shifting changes location but not spread; scaling by a scales distances by $|a|$, so squared distances scale by a^2 .

7.2 Independent sums

In general,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If X, Y are independent, covariance is zero and variances add. Thus for independent identically distributed variables with variance σ^2 ,

$$\text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{\sigma^2}{n}.$$

This $1/n$ variance and $1/\sqrt{n}$ standard-error scaling drives statistical precision.

7.3 Standard results

$$\begin{aligned}\text{Var}(\text{Bern}(p)) &= p(1-p), & \text{Var}(\text{Bin}(n, p)) &= np(1-p), \\ \text{Var}(\text{Geom}(p)) &= \frac{1-p}{p^2}, & \text{Var}(\text{Pois}(\lambda)) &= \lambda.\end{aligned}$$

Worked example

If individual delivery times have standard deviation 12 minutes and 100 independent deliveries are averaged, the standard deviation of the sample mean is $12/\sqrt{100} = 1.2$ minutes. Averaging removes noise slowly: to cut standard error by a factor of 10, you need 100 times as much data.

Exercises

1. If $\mathbb{E}X = 3$ and $\mathbb{E}X^2 = 14$, find $\text{Var}(X)$.
2. Prove that $\text{Var}(X) \geq 0$ and explain when equality holds.
3. If $X_1, \dots, X_{25}$ are independent with variance 9, find the variance and standard deviation of their average.
4. A Bernoulli variance is maximized at what value of p ? Interpret the result.

Selected solutions

1. $14 - 9 = 5$. 2. It is an expectation of a nonnegative quantity; equality means $(X - \mathbb{E}X)^2 = 0$ with probability 1, so X is constant almost surely. 3. Variance $9/25 = 0.36$, standard deviation 0.6. 4. $p(1-p)$ is maximized at $p = 1/2$: uncertainty is greatest when success and failure are equally plausible.

Continuous Random Variables: PDFs, CDFs, and Probability as Area

Intuition

For a continuous variable, probability is spread over an interval rather than concentrated on individual points. A probability density is therefore not a point probability. It is better imagined as “probability per unit of x .” Probability is the area under the density curve.

8.1 Density and distribution function

A continuous random variable has a density f_X satisfying

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

For $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

The CDF is

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

Where differentiable, $F'_X(x) = f_X(x)$.

Formal statement

For a genuinely continuous distribution, $\mathbb{P}(X = x) = 0$ for every single point. This does not mean the point is impossible in an everyday sense. It means a point has zero width and therefore zero area under the density.

8.2 Density can exceed one

A density is not a probability. For $X \sim \text{Unif}(0, 0.1)$, $f(x) = 10$ on that interval. The total area is still $10 \times 0.1 = 1$. Units help: if x is measured in seconds, density has units “per second,” so integrating over seconds produces a dimensionless probability.

8.3 Transformation by a monotone function

If $Y = g(X)$ and g is differentiable and strictly monotone, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

The derivative factor stretches or compresses density so that total probability is preserved.

Proof / derivation

For increasing g ,

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiate with the chain rule. For decreasing g , the same calculation introduces an absolute value.

Worked example

Suppose $X \sim \text{Unif}(0, 1)$ and $Y = X^2$. For $0 < y < 1$, $g^{-1}(y) = \sqrt{y}$ and $d\sqrt{y}/dy = 1/(2\sqrt{y})$. Thus

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1.$$

The density piles up near zero because squaring compresses values near zero more strongly.

Exercises

1. Let $f(x) = cx^2$ on $0 \leq x \leq 2$. Find c and $\mathbb{P}(X > 1)$.
2. If $F(x) = 1 - e^{-2x}$ for $x \geq 0$, find the density and $\mathbb{P}(1 < X < 2)$.
3. Let $X \sim \text{Unif}(0, 1)$ and $Y = -\log X$. Derive the density of Y .
4. Explain in words why a narrow interval can have small probability even if the density is large there.

Selected solutions

1. $1 = c \int_0^2 x^2 dx = 8c/3$, so $c = 3/8$; $\mathbb{P}(X > 1) = \frac{3}{8} \int_1^2 x^2 dx = 7/8$.
2. $f(x) = 2e^{-2x}$ and the probability is $F(2) - F(1) = e^{-2} - e^{-4}$.
3. Since $X = e^{-Y}$, $f_Y(y) = e^{-y}$ for $y \geq 0$, an exponential distribution with rate 1.
4. Probability is density times width in the local approximation; high height can be offset by tiny width.

Uniform, Exponential, and Normal Distributions

Intuition

These three continuous distributions play three different roles. Uniform represents “all locations in an interval are equally plausible.” Exponential models waiting times with no memory. Normal appears when many small independent effects add together and is therefore the central approximation distribution of statistics.

9.1 Uniform distribution

If $X \sim \text{Unif}(a, b)$,

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

Its mean and variance are

$$\mathbb{E}X = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

9.2 Exponential distribution

If $X \sim \text{Exp}(\lambda)$ with rate $\lambda > 0$,

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad F(x) = 1 - e^{-\lambda x}.$$

The survival function is $\mathbb{P}(X > x) = e^{-\lambda x}$, so

$$\mathbb{P}(X > s+t \mid X > s) = e^{-\lambda t}.$$

Thus exponential waiting time is the continuous memoryless distribution. Its mean is $1/\lambda$ and variance is $1/\lambda^2$.

9.3 Normal distribution

If $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardizing gives

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Then probabilities reduce to the standard normal CDF Φ : $\mathbb{P}(X \leq x) = \Phi((x - \mu)/\sigma)$.

Why the normalizing constant is $1/\sqrt{2\pi}$

Let $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx$. Then

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy.$$

Switch to polar coordinates: $x^2 + y^2 = r^2$ and $dx dy = r dr d\theta$. Hence

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi.$$

So $I = \sqrt{2\pi}$.

Worked example

A manufacturing measurement is approximately normal with mean 50 mm and standard deviation 2 mm. The probability of exceeding 53 mm is

$$1 - \Phi\left(\frac{53 - 50}{2}\right) = 1 - \Phi(1.5) \approx 0.0668.$$

The same standardization works for any normal mean and scale.

Common pitfall

“Normal” does not mean “typical data.” Many real quantities are skewed, heavy-tailed, bounded, zero-inflated, or multimodal. The normal model is powerful because of its mathematical structure and the central limit theorem, not because every histogram is bell-shaped.

Exercises

1. If $X \sim \text{Unif}(3, 9)$, find $\mathbb{P}(4 < X < 7)$, $\mathbb{E}X$, and $\text{Var } X$.
2. An exponential waiting time has mean 5 minutes. Find the rate and the probability of waiting more than 10 minutes.
3. If $X \sim \mathcal{N}(100, 15^2)$, express $\mathbb{P}(85 < X < 130)$ using Φ .
4. Prove the memoryless property of the exponential distribution using its survival function.

Selected solutions

1. $3/6 = 1/2$, mean 6, variance $36/12 = 3$.
2. $\lambda = 1/5$; probability e^{-2} .
3. $\Phi(2) - \Phi(-1)$.
4. $\mathbb{P}(X > s + t \mid X > s) = e^{-\lambda(s+t)} / e^{-\lambda s} = e^{-\lambda t}$.

Continuous Expectation, Variance, Quantiles, and Histograms

Intuition

The formulas for continuous variables are the same ideas as in the discrete case, with sums replaced by integrals. A histogram is the empirical counterpart of a density: it shows where observed data concentrate. Quantiles slice a distribution by probability mass rather than by physical width.

10.1 Expectation and LOTUS

For density f_X ,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

Variance is again $\mathbb{E}[X^2] - (\mathbb{E}X)^2$.

10.2 Tail-integral formula

For a nonnegative continuous X ,

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X > t) dt.$$

This identity says the mean equals the area under the survival curve.

Proof / derivation

For $x \geq 0$, write $x = \int_0^{\infty} \mathbf{1}\{t < x\} dt$. Therefore

$$\mathbb{E}[X] = \mathbb{E} \left[\int_0^{\infty} \mathbf{1}\{t < X\} dt \right] = \int_0^{\infty} \mathbb{P}(X > t) dt,$$

where swapping expectation and integral is justified for a nonnegative integrand.

10.3 Quantiles and robust summaries

The q th quantile x_q satisfies $F(x_q) = q$ for a strictly increasing continuous CDF. The interquartile range is $Q_{0.75} - Q_{0.25}$. Unlike the variance, it is resistant to extreme outliers.

10.4 Histograms

A density histogram divides the line into bins and chooses each bar height so that *bar area* equals the fraction of data in that bin. If bin widths differ, raw counts can be visually misleading; density normalization corrects this.

Worked example

For $X \sim \text{Exp}(\lambda)$, the tail formula gives

$$\mathbb{E}X = \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda},$$

a one-line derivation that is often simpler than integrating $x\lambda e^{-\lambda x}$.

Exercises

1. Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ for $X \sim \text{Unif}(0, 1)$.
2. Find the median of $\text{Exp}(\lambda)$.
3. For a normal distribution, explain why the mean and median coincide without doing an integral.
4. A histogram uses bins of widths 1 and 2 containing 20% and 30% of the observations. What should their density heights be?

Selected solutions

1. $1/2$ and $1/3$, so variance $1/12$.
2. Solve $1 - e^{-\lambda m} = 1/2$, giving $m = (\log 2)/\lambda$.
3. Symmetry around μ puts half the probability on either side, and symmetry also makes the balance point equal μ .
4. Heights are $0.2/1 = 0.2$ and $0.3/2 = 0.15$.

Manipulating and Combining Random Variables

Intuition

Real systems rarely produce a single primitive random variable. We add costs, average measurements, take maxima, convert units, or transform waiting times. The central skill is to move uncertainty through a function without losing track of the distribution.

11.1 Linear transformations

If $Y = aX + b$, then

$$\mathbb{E}Y = a\mathbb{E}X + b, \quad \text{Var}(Y) = a^2 \text{Var}(X).$$

For a density and $a \neq 0$,

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

11.2 Sums and convolution

For independent discrete X, Y ,

$$p_{X+Y}(z) = \sum_x p_X(x) p_Y(z-x).$$

For independent continuous variables,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

This is convolution. It expresses all ways the components can sum to the same total.

11.3 Maxima and minima

For independent $X_1, \dots, X_n$ with common CDF F ,

$$\mathbb{P}(\max_i X_i \leq x) = F(x)^n.$$

Similarly,

$$\mathbb{P}(\min_i X_i > x) = [1 - F(x)]^n.$$

These formulas are often easier than deriving densities directly.

Worked example

If three independent component lifetimes are exponential with rate λ and the system fails when the first component fails, then

$$\mathbb{P}(T_{\min} > t) = e^{-\lambda t} e^{-\lambda t} e^{-\lambda t} = e^{-3\lambda t}.$$

Thus the minimum is exponential with rate 3λ .

Exercises

1. If $X \sim \mathcal{N}(10, 4)$, find the distribution of $Y = 3X - 5$.
2. Two independent fair dice have sum S . Derive $\mathbb{P}(S = 8)$ by convolution.
3. If $X_1, \dots, X_5 \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$, find the CDF and density of $M = \max X_i$.
4. Two independent exponential variables have rates 2 and 3. Find the distribution of their minimum.

Selected solutions

1. $Y \sim \mathcal{N}(25, 36)$.
2. Sum over $x = 2, \dots, 6$: five valid pairs, so $5/36$.
3. $F_M(m) = m^5$ for $0 \leq m \leq 1$ and $f_M(m) = 5m^4$.
4. The survival function is $e^{-2t}e^{-3t} = e^{-5t}$, so the minimum is $\text{Exp}(5)$.

Law of Large Numbers and Central Limit Theorem

Intuition

Two different miracles happen when we average independent data. The law of large numbers says the average stabilizes near the true mean. The central limit theorem says the remaining fluctuations, after the right rescaling, become approximately normal. LLN explains consistency; CLT explains error bars.

12.1 Weak law of large numbers

Let $X_1, X_2, \dots$ be iid with mean μ and finite variance σ^2 . The sample mean $\bar{X}_n = n^{-1} \sum_i X_i$ satisfies

$$\bar{X}_n \xrightarrow{P} \mu,$$

meaning for every $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0.$$

Proof via Chebyshev

Because the variables are independent,

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Chebyshev's inequality gives

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

This proves the weak law under finite variance.

12.2 Central limit theorem

Under standard iid assumptions with finite mean μ and variance $\sigma^2 > 0$,

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Hence, for large n ,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

CLT proof sketch via MGFs

Assume the moment-generating function exists near zero. Standardize $Y_i = (X_i - \mu)/\sigma$, so $\mathbb{E}Y_i = 0$ and $\text{Var } Y_i = 1$. A Taylor expansion of its MGF gives

$$M_Y(t) = 1 + \frac{t^2}{2} + o(t^2).$$

For $S_n = (Y_1 + \dots + Y_n)/\sqrt{n}$, independence gives

$$M_{S_n}(t) = \left[M_Y\left(\frac{t}{\sqrt{n}}\right) \right]^n = \left[1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right]^n \rightarrow e^{t^2/2},$$

which is the MGF of $\mathcal{N}(0, 1)$. A fully general proof uses characteristic functions because MGFs need not exist.

12.3 Normal approximation to binomial

If $X \sim \text{Bin}(n, p)$ and $np, n(1-p)$ are sufficiently large,

$$X \approx \mathcal{N}(np, np(1-p)).$$

For integer counts, a continuity correction often improves accuracy, e.g. $\mathbb{P}(X \leq k) \approx \mathbb{P}(Y \leq k + 0.5)$.

Worked example

A website has conversion probability $p = 0.2$. For $n = 1000$ visitors, the sample conversion rate $\hat{p}$ has mean 0.2 and standard error

$$\sqrt{\frac{0.2(0.8)}{1000}} \approx 0.01265.$$

By the CLT, $\hat{p}$ is approximately normal. The natural scale of random fluctuation is about 1.3 percentage points.

Common pitfall

The CLT concerns properly standardized *sums or averages*; it does not say the raw data become normal. Also, heavy tails or strong dependence can make normal approximation slow or invalid.

Exercises

1. If iid observations have mean 50 and variance 100, what are the mean and standard deviation of the average of 64 observations?
2. Use Chebyshev to bound $\mathbb{P}(|\bar{X} - 50| \geq 5)$ for the sample in problem 1.
3. Approximate $\mathbb{P}(X \geq 60)$ for $X \sim \text{Bin}(100, 0.5)$ using a normal approximation with continuity correction.
4. Explain the conceptual difference between LLN and CLT in one sentence each.

Selected solutions

1. Mean 50, standard deviation $10/8 = 1.25$. 2. At most $100/(64 \cdot 25) = 0.0625$. 3. Use $Y \sim \mathcal{N}(50, 25)$: $\mathbb{P}(X \geq 60) \approx \mathbb{P}(Y \geq 59.5) = 1 - \Phi(1.9)$. 4. LLN: averages concentrate at the population mean. CLT: the shape of their scaled error approaches a normal law.

Joint Distributions, Marginals, and Conditional Distributions

Intuition

A single variable tells one story. A joint distribution tells how several stories move together. Once the joint law is known, marginals are obtained by “forgetting” variables, while conditional distributions are obtained by fixing information and renormalizing.

13.1 Discrete joint PMF

For discrete X, Y ,

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

Marginals are

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Conditional PMFs are

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Independence means $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for every pair.

13.2 Continuous joint density

A joint density $f_{X,Y}$ satisfies

$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y) dx dy = 1.$$

Probability is volume over a region A :

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy.$$

Marginals integrate out the other coordinate:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

13.3 Conditional expectation

The conditional mean $\mathbb{E}[X | Y = y]$ is the mean under the conditional distribution. As y varies it becomes a function of Y , written $\mathbb{E}[X | Y]$. The tower property says

$$\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X].$$

This is a probability version of “average within groups, then average the group averages with the right weights.”

Tower property, discrete case

$$\sum_y \mathbb{E}[X | Y = y] p_Y(y) = \sum_y \sum_x x p_{X|Y}(x | y) p_Y(y) = \sum_{x,y} x p_{X,Y}(x, y) = \mathbb{E}[X].$$

Worked example

A store’s daily demand D depends on weather W . If sunny days occur with probability 0.7 and have mean demand 100, while rainy days occur with probability 0.3 and have mean demand 160, then

$$\mathbb{E}[D] = \mathbb{E}[\mathbb{E}(D | W)] = 0.7(100) + 0.3(160) = 118.$$

The tower property lets us solve a mixture model without deriving its full distribution.

Exercises

1. A joint PMF has values proportional to $x + y$ for $x, y \in \{1, 2\}$. Normalize it and find both marginals.
2. If X, Y are independent, show $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ when expectations exist.
3. A joint density is $f(x, y) = 2$ on $0 < y < x < 1$ and 0 elsewhere. Find $f_X(x)$ and $f_Y(y)$.
4. Explain why zero covariance will later be weaker than independence.

Selected solutions

1. Weights are 2,3,3,4, total 12. Hence e.g. $p_X(1) = 5/12$, $p_X(2) = 7/12$, and symmetry gives the same for Y .
2. Factor the double sum or integral using $p_{X,Y} = p_X p_Y$.
3. $f_X(x) = \int_0^x 2dy = 2x$; $f_Y(y) = \int_y^1 2dx = 2(1 - y)$.
4. Covariance measures only linear co-movement; nonlinear dependence can average to zero covariance.

Covariance, Correlation, and Dependence

Intuition

Variance measures one variable's spread. Covariance measures whether two variables tend to be above their means together. Correlation rescales covariance into a unit-free number between -1 and 1. Neither quantity captures every kind of dependence.

14.1 Definitions

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X \mathbb{E}Y.$$

If $\sigma_X, \sigma_Y > 0$,

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

14.2 Variance of a sum

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

This explains why dependence changes uncertainty. Positive covariance makes sums more volatile; negative covariance can create diversification.

14.3 Independence implies zero covariance

If X, Y are independent and have finite second moments, then

$$\mathbb{E}[XY] = \mathbb{E}X \mathbb{E}Y,$$

so $\text{Cov}(X, Y) = 0$. The converse is false.

Worked example

Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2$. Then Y is completely determined by X , so the variables are dependent. But symmetry gives $\mathbb{E}X = 0$ and $\mathbb{E}[X^3] = 0$, hence

$$\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}X \mathbb{E}[X^2] = 0.$$

Correlation misses this U-shaped relationship.

14.4 Why correlation is between -1 and 1

Cauchy-Schwarz implies

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X) \text{Var}(Y)},$$

so $|\text{Corr}(X, Y)| \leq 1$. Equality occurs exactly when centered variables are linearly related almost surely.

Cauchy-Schwarz route

For centered $U = X - \mathbb{E}X$ and $V = Y - \mathbb{E}Y$, Cauchy-Schwarz says $|\mathbb{E}[UV]|^2 \leq \mathbb{E}[U^2]\mathbb{E}[V^2]$. Replace the terms by covariance and variances and divide by $\sigma_X^2 \sigma_Y^2$.

Exercises

1. If $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, and $\text{Cov}(X, Y) = 3$, find $\text{Var}(2X - Y)$.
2. Show $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$.
3. Give a real-world example where two quantities are correlated because of a common cause rather than one causing the other.
4. If $\text{Corr}(X, Y) = -1$, what precise structural relationship must hold between X and Y ?

Selected solutions

1. $4(4) + 9 - 4(3) = 13$.
2. Constants disappear after centering and multiplicative constants factor out.
3. Example: ice-cream sales and sunburn incidence both increase with hot weather.
4. There exist constants $a < 0, b$ such that $Y = aX + b$ almost surely, assuming nonzero variances.

Part II

Statistics I: Learning from Data with Bayesian Inference

From Probability to Statistics: Data, Models, Likelihood, and MLE

Intuition

Probability moves forward: a model and parameter generate possible data. Statistics runs the movie backward: we observe data and ask which parameter values make those data plausible. The likelihood function is the bridge. It uses the same formula as a probability model, but treats the data as fixed and the parameter as the variable.

15.1 Data, parameter, statistic

A statistical model is a family $\{p(x | \theta) : \theta \in \Theta\}$ or $\{f(x | \theta)\}$. The unknown θ is a parameter. A statistic $T(X_1, \dots, X_n)$ is any function of the observed sample that does not depend on unknown parameters.

15.2 Probability versus likelihood

For fixed θ , $p(x | \theta)$ is a probability distribution over possible x . After observing x , the same expression viewed as a function of θ is the likelihood:

$$L(\theta; x) = p(x | \theta).$$

A likelihood need not integrate or sum to one over θ .

15.3 Maximum likelihood estimation

The MLE is

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta; x).$$

Products are often simplified using the log-likelihood $\ell(\theta) = \log L(\theta)$; logarithms preserve the maximizer and turn products into sums.

Worked example

For iid Bernoulli data $x_1, \dots, x_n \in \{0, 1\}$ with success probability p ,

$$L(p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}.$$

Let $s = \sum x_i$. Then

$$\ell(p) = s \log p + (n - s) \log(1 - p).$$

Differentiating and setting to zero gives

$$\frac{s}{p} - \frac{n - s}{1 - p} = 0 \quad \Rightarrow \quad \hat{p} = \frac{s}{n} = \bar{x}.$$

The MLE is the sample success rate.

15.4 MLEs for familiar models

For iid $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with known σ , $\hat{\mu} = \bar{X}$. For iid exponential data with rate λ , $\hat{\lambda} = 1/\bar{X}$. For Poisson counts, $\hat{\lambda} = \bar{X}$.

15.5 Exploratory plots

Histograms reveal shape, skew, and multiple modes. Scatter plots reveal association, nonlinear patterns, clusters, and outliers. A statistical model should be challenged by the data visualization before sophisticated inference is trusted.

Exercises

1. Derive the MLE of λ from iid Poisson observations.
2. Derive the MLE of the rate of an exponential sample.
3. Explain why multiplying a likelihood by a positive constant independent of θ does not change the MLE.
4. Give one reason a model can fit the mean of data well but still be badly wrong.

Selected solutions

1. $\ell(\lambda) = -n\lambda + (\sum x_i) \log \lambda + \text{const}$, so $-n + (\sum x_i)/\lambda = 0$, yielding $\hat{\lambda} = \bar{x}$.
2. $\ell(\lambda) = n \log \lambda - \lambda \sum x_i$, hence $\hat{\lambda} = n/\sum x_i = 1/\bar{x}$.
3. A positive constant adds a constant to log-likelihood and does not change where it is maximized.
4. The variance, tails, skew, dependence, or multimodality can be wrong even if the mean matches.

Bayesian Updating with Discrete Priors

Intuition

Bayesian inference treats uncertainty about an unknown parameter using a probability distribution. Before data, we have a prior. The data contribute a likelihood. Multiplying prior by likelihood and renormalizing gives the posterior. This is Bayes' theorem applied to unknown hypotheses.

16.1 Discrete parameter space

Suppose θ can take values $\theta_1, \dots, \theta_k$. Let prior probabilities be $\pi_i = \mathbb{P}(\theta = \theta_i)$. After data x ,

$$\mathbb{P}(\theta_i | x) = \frac{p(x | \theta_i) \pi_i}{\sum_{j=1}^k p(x | \theta_j) \pi_j}.$$

In shorthand,

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

16.2 Sequential updating

If data arrive in stages, today's posterior can be tomorrow's prior. For conditionally independent observations,

$$\pi(\theta | x_1, \dots, x_n) \propto \pi(\theta) \prod_{i=1}^n p(x_i | \theta).$$

The order of data does not matter because multiplication commutes.

Worked example

A factory machine is either well calibrated (H_1) with defect probability 0.01 or poorly calibrated (H_2) with defect probability 0.08. Prior probabilities are 0.8 and 0.2. We inspect 5 independent items and see one defect. Ignoring the common combinatorial factor,

$$L(H_1) \propto 0.01(0.99)^4, \quad L(H_2) \propto 0.08(0.92)^4.$$

Posterior weights are these likelihoods times 0.8 and 0.2, then normalized. Even a single defect can shift belief toward the poor-calibration state, but how much depends on the prior and the likelihood ratio.

16.3 Posterior summaries

Once we have a posterior, we can report posterior probabilities, posterior mean $\mathbb{E}[\theta \mid x]$, posterior mode (MAP), or decision-relevant probabilities such as $\mathbb{P}(\theta > \theta_0 \mid x)$. There is no rule saying one summary is always best; the right summary depends on the decision or scientific question.

Common pitfall

A prior is not the same as “bias.” It is part of the model. A strong prior can dominate little data, while a weak prior allows the likelihood to dominate. The scientific issue is whether the prior accurately represents information available before the current data.

Exercises

1. A coin is either fair ($p = 0.5$) or biased ($p = 0.8$), each with prior probability 0.5. After observing HHTH, find the posterior probability the coin is biased.
2. Repeat problem 1 with prior probability 0.1 for the biased coin. What changes conceptually?
3. Show algebraically that updating on x_1 then x_2 gives the same posterior as updating on (x_1, x_2) at once when observations are conditionally independent.
4. Give an example where a discrete prior is more natural than a continuous prior.

Selected solutions

1. Likelihoods are $0.5^4 = 0.0625$ and $0.8^3(0.2) = 0.1024$. Posterior biased probability is $0.1024 / (0.1024 + 0.0625) \approx 0.621$.
2. Normalize $0.1(0.1024)$ against $0.9(0.0625)$, giving about 0.154; the same evidence creates a smaller posterior because prior odds were much lower.
3. Multiplying by $p(x_1 \mid \theta)$ and then $p(x_2 \mid \theta)$ gives the same product.
4. A machine known to be one of three hardware revisions is a discrete latent state.

Bayesian Prediction, Posterior Odds, and Evidence

Intuition

Inference about a parameter is often only an intermediate goal. Usually we care about what happens next. Bayesian prediction averages the future-data model over our posterior uncertainty about the parameter. Posterior odds give a compact way to see how evidence changes competing hypotheses.

17.1 Posterior predictive distribution

For future observation X_{new} ,

$$p(x_{\text{new}} | x) = \sum_{\theta} p(x_{\text{new}} | \theta) \mathbb{P}(\theta | x)$$

for a discrete parameter, or

$$p(x_{\text{new}} | x) = \int p(x_{\text{new}} | \theta) \pi(\theta | x) d\theta$$

for a continuous parameter. Prediction averages over parameter uncertainty rather than plugging in a single estimate.

17.2 Odds and Bayes factors

For hypotheses H_1, H_0 ,

$$\frac{\mathbb{P}(H_1 | x)}{\mathbb{P}(H_0 | x)} = \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \frac{p(x | H_1)}{p(x | H_0)}.$$

The likelihood ratio or Bayes factor measures how much the data alone shift the odds.

Worked example

A diagnostic hypothesis H_1 has prior odds 1 : 19. A result has likelihood ratio 10 in favor of H_1 . Posterior odds are 10 : 19, giving posterior probability $10/(10 + 19) \approx 0.345$. Strong evidence may still leave the posterior below 50% if the hypothesis was initially rare.

17.3 Prediction versus plug-in estimation

If p is uncertain and $X_{\text{new}} \sim \text{Bern}(p)$, the plug-in predictor uses $\hat{p}$. Bayesian prediction uses $\mathbb{E}[p \mid x]$ for one Bernoulli trial. For nonlinear future quantities, integrating the full posterior can differ substantially from evaluating at the posterior mean.

Predictive probability for a Bernoulli future

$$\mathbb{P}(X_{\text{new}} = 1 \mid x) = \int \mathbb{P}(X_{\text{new}} = 1 \mid p) \pi(p \mid x) dp = \int p \pi(p \mid x) dp = \mathbb{E}[p \mid x].$$

Exercises

1. Prior odds for H_1 are 1 : 4. Evidence has likelihood ratio 6. Find posterior probability of H_1 .
2. Under a discrete posterior $\mathbb{P}(p = 0.2 \mid x) = 0.3$, $\mathbb{P}(p = 0.7 \mid x) = 0.7$, find the predictive probability the next Bernoulli trial succeeds.
3. Explain why posterior predictive uncertainty is usually larger than uncertainty computed as if the parameter were known exactly.
4. Two independent evidence items have likelihood ratios 3 and 5. What is their combined effect on prior odds?

Selected solutions

1. Posterior odds are $6 : 4 = 3 : 2$, so probability is $3/5 = 0.6$.
2. $0.3(0.2) + 0.7(0.7) = 0.55$.
3. Prediction includes both randomness of the future observation and uncertainty about the unknown parameter.
4. Multiply: combined likelihood ratio is 15.

Bayesian Updating with Continuous Priors

Intuition

When a parameter can take infinitely many values, posterior probabilities at individual points are usually zero. We therefore work with densities. The logic is unchanged: posterior density is proportional to likelihood times prior density, and the denominator is simply the constant that makes the area equal one.

18.1 Bayes' rule for densities

For parameter density $\pi(\theta)$ and observed data x ,

$$\pi(\theta | x) = \frac{f(x | \theta)\pi(\theta)}{\int f(x | u)\pi(u)du}.$$

The denominator is called the marginal likelihood or evidence.

18.2 Kernel recognition

In practice we often write

$$\pi(\theta | x) \propto f(x | \theta)\pi(\theta)$$

and recognize the functional shape. Constants not involving θ can be dropped while deriving the posterior kernel.

Worked example

Suppose one Bernoulli success probability p has uniform prior on $[0, 1]$, and in n trials we see s successes and $f = n - s$ failures. Then

$$\pi(p | x) \propto p^s(1 - p)^f, \quad 0 < p < 1.$$

This is a $\text{Beta}(s + 1, f + 1)$ density. The posterior mean is

$$\frac{s + 1}{n + 2},$$

a smoothed version of the MLE s/n .

18.3 Continuous data with continuous priors

If $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2 and prior $\mu \sim \mathcal{N}(m_0, s_0^2)$, the posterior is also normal. The posterior precision is the sum of prior and data precisions:

$$\frac{1}{s_n^2} = \frac{1}{s_0^2} + \frac{n}{\sigma^2}.$$

The posterior mean is a precision-weighted average,

$$m_n = s_n^2 \left(\frac{m_0}{s_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

Normal-normal conjugacy

Ignoring constants independent of μ , the prior-likelihood product is

$$\exp \left[-\frac{(\mu - m_0)^2}{2s_0^2} - \frac{n(\mu - \bar{x})^2}{2\sigma^2} \right].$$

Collect the μ^2 and μ terms and complete the square. The coefficient of μ^2 is the sum of precisions, and the center of the square is the stated weighted mean. Thus the kernel is normal.

Exercises

1. With a uniform prior on p and data 7 successes, 3 failures, identify the posterior distribution and mean.
2. For the normal-normal model with $m_0 = 0$, $s_0 = 10$, known $\sigma = 2$, $n = 4$, and $\bar{x} = 3$, compute m_n and s_n^2 .
3. Why may we drop binomial coefficients when deriving the posterior for p ?
4. Explain why the denominator in continuous Bayes' rule is needed even though we often ignore it during algebra.

Selected solutions

1. Beta(8, 4) with mean $8/12 = 2/3$.
2. Precision $1/100 + 4/4 = 1.01$, so $s_n^2 = 1/1.01 \approx 0.9901$ and $m_n = s_n^2(0 + 3) \approx 2.9703$.
3. They do not depend on p , so they are part of the normalizing constant.
4. The posterior must integrate to one; the denominator supplies exactly that normalization.

Conjugate Priors: Beta-Binomial and Normal-Normal

Intuition

A conjugate prior is chosen so that posterior updating stays in the same distribution family. Conjugacy is algebraically convenient and often interpretable as adding pseudo-observations. It is not a law of nature; it is a modeling convenience that can be useful when its shape makes sense.

19.1 Beta distribution

For $a, b > 0$,

$$p \sim \text{Beta}(a, b), \quad \pi(p) = \frac{1}{B(a, b)} p^{a-1} (1-p)^{b-1}, \quad 0 < p < 1.$$

Its mean and variance are

$$\mathbb{E}p = \frac{a}{a+b}, \quad \text{Var}(p) = \frac{ab}{(a+b)^2(a+b+1)}.$$

19.2 Beta-Binomial updating

If $p \sim \text{Beta}(a, b)$ and the data contain s successes and f failures, then

$$p \mid x \sim \text{Beta}(a+s, b+f).$$

Proof / derivation

The likelihood kernel is $p^s(1-p)^f$. Multiplying by the prior kernel gives

$$p^{a+s-1}(1-p)^{b+f-1},$$

which is the kernel of a $\text{Beta}(a+s, b+f)$ density.

19.3 Pseudo-count interpretation

The parameters a, b act roughly like prior success/failure counts, though the exact pseudo-count interpretation depends on which posterior summary you use. The posterior mean is

$$\frac{a + s}{a + b + n} = \frac{a + b}{a + b + n} \frac{a}{a + b} + \frac{n}{a + b + n} \frac{s}{n},$$

a weighted average of prior mean and sample proportion.

19.4 Normal-normal precision weighting

For the normal mean problem with known variance, conjugacy produces posterior precision

$$\tau_n = \tau_0 + \tau_{\text{data}},$$

where precision is inverse variance. Independent information adds on the precision scale.

Worked example

A prior $\text{Beta}(2, 2)$ expresses mild belief that a conversion rate is not extreme. After 8 successes and 12 failures, the posterior is $\text{Beta}(10, 14)$ with mean $10/24 \approx 0.417$. The MLE is $8/20 = 0.4$; the prior pulls the posterior mean slightly toward 0.5.

Exercises

1. Under prior $\text{Beta}(3, 7)$, observe 12 successes and 8 failures. Find posterior distribution and posterior mean.
2. Show algebraically that the Beta posterior mean is a weighted average of the prior mean and the sample proportion.
3. In the normal-normal model, what happens to the posterior mean when $s_0^2 \rightarrow \infty$?
4. What is one danger of choosing a conjugate prior solely because the algebra is easy?

Selected solutions

1. $\text{Beta}(15, 15)$ with mean $1/2$.
2. Expand the weighted-average formula above.
3. Prior precision goes to zero, so the posterior mean approaches $\bar{x}$ and posterior variance approaches σ^2/n .
4. The chosen family may encode unrealistic prior beliefs or tail behavior.

Choosing Priors and Sensitivity Analysis

Intuition

A prior should encode information available before the current data, but prior choice is rarely a single objectively correct number. A good Bayesian analysis asks how strongly the posterior depends on plausible prior alternatives. Sensitivity analysis is the statistical equivalent of stress-testing an assumption.

20.1 Informative, weakly informative, and reference-like priors

An informative prior concentrates on values supported by real prior knowledge. A weakly informative prior rules out absurd scales without forcing a narrow answer. A very diffuse prior tries to contribute little information, but “flat” depends on parameterization: a prior flat in θ is generally not flat in $g(\theta)$.

20.2 Prior predictive checking

Before seeing data x , the model implies a prior predictive distribution

$$p(x) = \int p(x | \theta) \pi(\theta) d\theta.$$

Simulating from this distribution asks: does the prior-plus-likelihood model generate datasets that look remotely plausible? If it routinely predicts impossible values, the prior/model pair needs revision.

20.3 Sensitivity

Repeat the analysis under several scientifically defensible priors. If conclusions barely move, the data dominate. If conclusions move dramatically, report that honestly: the available data do not identify the answer strongly without prior assumptions.

Worked example

Suppose an A/B test sees 3 conversions in 10 visits. Under Beta(1, 1) the posterior is Beta(4, 8); under Beta(20, 20) it is Beta(23, 27). The second prior acts like much more prior information near 0.5, so the same small dataset moves it less. The point is not that one prior is “wrong”; the analyst must justify which information state is realistic.

Exercises

1. Compare posterior means after 5 successes in 10 trials under $\text{Beta}(1, 1)$ and $\text{Beta}(100, 100)$ priors.
2. Why is a uniform prior on a probability p not uniform on the log-odds $\log(p/(1 - p))$?
3. Describe a prior predictive check for daily customer counts under a Poisson-rate model.
4. What conclusion should you report if a key posterior probability ranges from 0.55 to 0.95 over reasonable priors?

Selected solutions

1. Means are $6/12 = 0.5$ and $105/210 = 0.5$ here; although means coincide because data are exactly balanced, posterior variances differ greatly.
2. Probability density changes under nonlinear transformations via the Jacobian.
3. Draw λ from the prior, then draw Poisson counts; compare simulated ranges, zeros, extremes, and variability with domain knowledge.
4. The substantive conclusion is prior-sensitive; do not present one posterior number as robust.

Bayesian Probability Intervals and Credible Intervals

Intuition

A Bayesian interval is a direct probability statement about an unknown parameter under the posterior model. A 95% credible interval contains 95% of posterior probability. This interpretation is different from the frequentist confidence-interval interpretation studied later.

21.1 Equal-tail intervals

For posterior CDF $F(\theta | x)$, a central 95% interval is

$$[F^{-1}(0.025), F^{-1}(0.975)].$$

It leaves 2.5% posterior probability in each tail.

21.2 Highest posterior density intervals

An HPD region collects parameter values with the highest posterior density until the desired probability mass is reached. For a symmetric unimodal posterior, HPD and equal-tail intervals coincide; for skewed distributions they can differ.

21.3 Decision-focused probabilities

Sometimes an interval is less useful than a directly relevant posterior probability such as

$$\mathbb{P}(p_B - p_A > 0.02 \mid \text{data}).$$

Bayesian inference naturally supports such questions because uncertainty about parameters is represented probabilistically.

Worked example

After observing binomial data, suppose the posterior for p is $\text{Beta}(30, 20)$. The posterior mean is 0.6. A 95% credible interval is formed from the 2.5% and 97.5% beta quantiles. Its correct reading is: conditional on the model, prior, and observed data, 95% of posterior probability lies inside this interval.

Common pitfall

A 95% credible interval is not automatically a 95% frequentist confidence interval, and a 95% confidence interval is not usually allowed the posterior-probability interpretation. The numerical intervals can sometimes be similar while the logic differs.

Exercises

1. Distinguish an equal-tail interval from an HPD interval.
2. Why can a posterior interval be asymmetric around the posterior mean?
3. For a posterior $\mathcal{N}(10, 2^2)$, write the approximate central 95% credible interval.
4. Give a decision question that might be better answered by a posterior tail probability than by an interval.

Selected solutions

1. Equal-tail allocates equal probability to both tails; HPD contains the densest values and need not split tail probability equally.
2. The posterior distribution itself may be skewed.
3. Approximately $10 \pm 1.96(2) = [6.08, 13.92]$.
4. Example: probability a new treatment improves success probability by at least 5 percentage points.

Part III

Statistics II: Frequentist Inference and Hypothesis Testing

The Frequentist Framework and Sampling Distributions

Intuition

Frequentist inference treats the parameter as fixed but unknown and the data as random. The key object is the sampling distribution: what would our statistic do across repeated datasets generated under the same true parameter? Standard errors, p-values, and confidence intervals all come from this repeated-sampling viewpoint.

22.1 Estimator and sampling distribution

An estimator $\hat{\theta} = T(X_1, \dots, X_n)$ is random before data are collected. Its sampling distribution is the distribution of T under repeated samples. Important properties include

$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta,$$

variance, mean-squared error

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{bias}(\hat{\theta})^2,$$

and consistency.

22.2 Standard error

The standard error of an estimator is the standard deviation of its sampling distribution. For iid data with variance σ^2 ,

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

If σ is unknown, it is commonly estimated by the sample standard deviation s .

22.3 Unbiased sample variance

The sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The $n-1$ denominator corrects the downward bias created by estimating the center with the same data.

Why $n - 1$ appears

Use the identity

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations gives $n\sigma^2 - n(\sigma^2/n) = (n - 1)\sigma^2$. Dividing by $n - 1$ makes s^2 unbiased.

Worked example

If a process has standard deviation 20 and we average $n = 100$ independent observations, the standard error is 2. If we average $n = 400$, it is 1. Quadrupling data only halves standard error, a central fact in experimental design.

Exercises

1. Define the difference between the population standard deviation and the standard error of the sample mean.
2. An estimator has bias 2 and variance 9. Find its MSE.
3. Show that $\bar{X}$ is unbiased for μ for iid observations with finite mean.
4. If you want to reduce standard error by a factor of 3, approximately how much must sample size increase?

Selected solutions

1. Population SD describes spread of individual observations; SE describes spread of an estimator across repeated samples.
2. $9 + 4 = 13$.
3. Linearity gives $\mathbb{E}\bar{X} = n^{-1} \sum \mathbb{E}X_i = \mu$.
4. By a factor of $3^2 = 9$.

Null Hypothesis Significance Testing, p-Values, and z-Tests

Intuition

A significance test is a stress test for a null model. We temporarily assume H_0 is true, ask how surprising our statistic would be under that assumption, and quantify the extremeness with a p-value. A small p-value is evidence against the null model; it is not the probability that the null is true.

23.1 Ingredients of a test

A hypothesis test specifies

- a null hypothesis H_0 and alternative H_1 ;
- a test statistic T whose distribution under H_0 is known or approximated;
- a rejection region or p-value rule;
- a significance level α , the target Type I error probability.

A Type I error rejects a true H_0 ; its probability is controlled by α . A Type II error fails to reject a false H_0 . Power is

$$\text{Power}(\theta) = \mathbb{P}_\theta(\text{reject } H_0),$$

so under alternatives it equals $1 - \beta(\theta)$.

23.2 p-values

For a right-tailed test with larger T more incompatible with H_0 ,

$$p = \mathbb{P}_{H_0}(T \geq T_{\text{obs}}).$$

A two-sided p-value counts outcomes at least as incompatible in both directions; for symmetric null distributions it is often twice a one-tail probability.

Common pitfall

A p-value is *not* $\mathbb{P}(H_0 \mid \text{data})$. It is computed assuming H_0 . It also is not the probability the result occurred “by chance” and does not measure effect size.

23.3 One-sample z-test

If $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with known σ , or the normal approximation is justified, test $H_0 : \mu = \mu_0$ using

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Under H_0 , $Z \sim \mathcal{N}(0, 1)$ exactly in the normal-known-variance model.

Worked example

A filling machine should average 500 g. Known process SD is 8 g. From $n = 64$ packages we observe $\bar{x} = 497.5$. Testing $H_0 : \mu = 500$ versus $H_1 : \mu \neq 500$ gives

$$Z = \frac{-2.5}{8/8} = -2.5.$$

The two-sided p-value is $2\Phi(-2.5) \approx 0.0124$, evidence against the 500 g null at the 5% level.

23.4 Rejection regions and p-values are equivalent views

At level $\alpha = 0.05$, a two-sided z-test rejects when $|Z| > 1.96$. Equivalently, reject when the p-value is below 0.05. The first predefines extreme values of the statistic; the second reports how far into the tail the observed statistic lies.

23.5 Power and sample size

Power increases with larger true effects, lower noise, larger sample size, and a more permissive significance threshold. Statistical significance with huge n can correspond to a practically tiny effect, so effect sizes and intervals should accompany tests.

Exercises

1. State Type I and Type II errors for testing whether a medical treatment has no effect.
2. A z-statistic equals 2.1 in a two-sided test. Express its p-value using Φ .
3. At $\alpha = 0.01$, what is the approximate two-sided z critical value?
4. Why can “fail to reject H_0 ” not be replaced by “accept H_0 ”?

Selected solutions

1. Type I: claim an effect when none exists. Type II: fail to detect an effect that exists. 2. $2[1 - \Phi(2.1)]$. 3. About 2.576 in absolute value. 4. A non-significant result may simply reflect low power; it does not prove the null exactly true.

t-Distributions and One- and Two-Sample t-Tests

Intuition

When the population standard deviation is unknown, replacing it by the sample standard deviation adds uncertainty. Student's t distribution is a normal distribution with extra tail mass that accounts for this estimated scale. As sample size grows, the scale estimate stabilizes and t approaches normal.

24.1 Student's t distribution

If $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_\nu^2$ independently, then

$$T = \frac{Z}{\sqrt{V/\nu}} \sim t_\nu.$$

The degrees of freedom ν control tail heaviness. For normal data,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

24.2 One-sample t-test

To test $H_0 : \mu = \mu_0$ when σ is unknown,

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

with $n - 1$ degrees of freedom under normal sampling. For moderate or large n , the procedure is often robust to modest non-normality.

24.3 Paired t-test

For matched observations (A_i, B_i) , define differences $D_i = A_i - B_i$ and perform a one-sample t-test on D_i . Pairing uses within-subject or within-unit comparisons and can remove large baseline variability.

24.4 Two independent samples

For groups with means μ_1, μ_2 , a robust default is Welch's t-statistic

$$T = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with Welch-Satterthwaite approximate degrees of freedom. It does not require equal population variances.

Worked example

Two versions of a service have mean response times 210 ms ($s_1 = 30, n_1 = 40$) and 195 ms ($s_2 = 25, n_2 = 35$). The estimated difference is 15 ms and the standard error is

$$\sqrt{30^2/40 + 25^2/35} \approx 6.35.$$

Thus $T \approx 2.36$. The inferential question is whether this magnitude is unusual under a zero-difference model, while the practical question is whether 15 ms matters operationally.

Why t appears, conceptually

For normal samples, $\bar{X}$ and S^2 are independent, $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ is standard normal, and $(n-1)S^2/\sigma^2$ is chi-square with $n-1$ degrees of freedom. Dividing the normal statistic by the square root of the scaled chi-square variable yields a t_{n-1} distribution.

Exercises

1. A sample has $n = 16$, $\bar{x} = 12$, $s = 4$. Write the t-statistic for testing $H_0 : \mu = 10$ and state its df.
2. When is a paired test preferable to an independent two-sample test?
3. Why is Welch's test generally safer than the equal-variance pooled t-test when variances differ?
4. What happens to the t_ν distribution as $\nu \rightarrow \infty$?

Selected solutions

1. $T = (12 - 10)/(4/4) = 2$ with 15 df.
2. When measurements are naturally matched, such as before/after on the same person, because analyzing differences captures the pairing.
3. It accounts for unequal variance without forcing a common variance estimate.
4. It converges to the standard normal distribution.

Chi-Square Distributions and Chi-Square Tests

Intuition

Chi-square statistics measure squared discrepancies. If observed categorical counts differ from expected counts, each discrepancy is scaled by its expected size, squared, and added. Under a suitable null model, that total has approximately a chi-square distribution.

25.1 Chi-square distribution

If $Z_1, \dots, Z_\nu$ are independent standard normal variables,

$$Q = \sum_{i=1}^{\nu} Z_i^2 \sim \chi_\nu^2.$$

It is nonnegative and right-skewed, becoming more symmetric as degrees of freedom grow.

25.2 Goodness-of-fit test

Suppose categorical counts O_i have null expected counts $E_i = np_i$. Pearson's statistic is

$$X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}.$$

If expected counts are sufficiently large and the null probabilities are fully specified, X^2 is approximately χ_{k-1}^2 .

25.3 Test of independence in a contingency table

For a table of counts, under independence the expected count in cell (i, j) is

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{n}.$$

Then

$$X^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

has approximately $(r-1)(c-1)$ degrees of freedom under the null.

Worked example

Suppose 200 users are classified by device (mobile/desktop) and purchase (yes/no). A chi-square independence test asks whether the pattern of purchase proportions differs beyond what random fluctuation would typically produce if device and purchase were independent. It tests association, not causation.

25.4 Connection to variance inference

For normal data,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

This result underlies classical confidence intervals and tests for a normal population variance.

Exercises

1. In a goodness-of-fit test with 6 fully specified categories, what are the df?
2. A 3×4 contingency table is tested for independence. What are the df?
3. Why do very small expected cell counts make the chi-square approximation unreliable?
4. If observed equals expected in every cell, what is Pearson's X^2 ?

Selected solutions

1. 5. 2. $(3-1)(4-1) = 6$. 3. The asymptotic approximation of discrete count fluctuations by the chi-square law is poor in sparse cells. 4. Zero.

One-Way ANOVA and the F Distribution

Intuition

ANOVA asks whether several group means can plausibly be equal by comparing two kinds of variation: variation of group means around the grand mean, and ordinary variation of observations within groups. If between-group variation is too large relative to within-group noise, equal means become implausible.

26.1 Model

For groups $j = 1, \dots, k$,

$$Y_{ij} = \mu_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

The null is

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k.$$

26.2 Variance decomposition

Let $\bar{y}_j$ be the group means and $\bar{y}$ the grand mean. Then

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}},$$

where

$$SS_{\text{Between}} = \sum_j n_j (\bar{y}_j - \bar{y})^2,$$

$$SS_{\text{Within}} = \sum_j \sum_i (y_{ij} - \bar{y}_j)^2.$$

The mean squares are

$$MS_B = \frac{SS_B}{k-1}, \quad MS_W = \frac{SS_W}{n-k}.$$

26.3 F statistic

Under H_0 and the classical model,

$$F = \frac{MS_B}{MS_W} \sim F_{k-1, n-k}.$$

Large values indicate group means are more spread out than within-group noise alone predicts.

Why a ratio of variances appears

Under the normal equal-mean null, suitable scaled versions of SS_B and SS_W are independent chi-square variables with $k - 1$ and $n - k$ degrees of freedom. The ratio of each chi-square divided by its df is, by definition, an F distribution. This is the multi-group analogue of comparing signal variance to noise variance.

Common pitfall

A significant ANOVA says at least one group mean differs; it does not identify which groups differ. Post-hoc comparisons require additional procedures and multiplicity control.

Exercises

1. For $k = 4$ groups and total $n = 40$, state ANOVA numerator and denominator df.
2. What does $F \approx 1$ suggest intuitively?
3. Name the main classical ANOVA assumptions.
4. Why are many separate pairwise t-tests without correction problematic?

Selected solutions

1. 3 and 36. 2. Between-group variability is about the same scale as within-group noise, giving little evidence of distinct means. 3. Independent observations/errors, normal errors within groups, and common variance in classical one-way ANOVA. 4. Repeated testing inflates the chance of at least one false positive.

Bayesian and Frequentist Inference: Two Logics

Intuition

Bayesian and frequentist methods answer different probability questions. Bayesian probability describes uncertainty about parameters conditional on the observed data and prior model. Frequentist probability describes how procedures behave over hypothetical repeated datasets with the parameter fixed. Many disagreements disappear once the conditioning direction is made explicit.

27.1 What is random?

	Bayesian	Frequentist
Parameter	represented by a prior/posterior distribution	fixed but unknown
Data	random before observation; fixed after observed	random in repeated-sampling calculations
Interval	posterior probability statement	procedure with long-run coverage
Test evidence	posterior odds, Bayes factors, posterior probabilities	test statistic, p-value, rejection rule

27.2 Neither logic removes modeling assumptions

Both schools rely on a likelihood or sampling model. Both can fail under bad measurement, selection bias, dependence, model misspecification, or data leakage. The philosophical framework cannot rescue poor data generation.

27.3 Why answers can resemble each other

With regular models, weak priors, and large samples, Bayesian posterior distributions and frequentist sampling approximations often concentrate near the same MLE with similar normal uncertainty. This is one reason familiar intervals can be numerically close even though their interpretations differ.

Worked example

For a large Bernoulli sample, a Beta prior has limited influence and the posterior for p is approximately normal around $\hat{p}$. A frequentist Wald-type confidence interval is also centered around $\hat{p}$. The numbers may nearly match, but “95% posterior probability” and “95% repeated-sampling coverage” are distinct statements.

Exercises

1. State the correct Bayesian and frequentist interpretations of a 95% interval.
2. Which framework naturally computes $\mathbb{P}(\theta > 0 \mid x)$ without introducing a prior?
3. Give two problems that can invalidate both Bayesian and frequentist conclusions.
4. Why can numerical agreement between methods be reassuring but not logically prove either interpretation?

Selected solutions

1. Bayesian: posterior probability 0.95 in the interval, conditional on model/prior/data. Frequentist: the interval-generating procedure covers the fixed parameter in 95% of repeated samples under assumptions.
2. Bayesian.
3. Examples: selection bias and dependence ignored by the model.
4. Similar approximations can produce similar numbers while conditioning on different hypothetical objects.

Part IV

Statistics III: Confidence Intervals, Bootstrap, and Regression

Confidence Intervals: Construction and Interpretation

Intuition

A confidence interval is a random interval procedure designed to catch a fixed unknown parameter at a stated long-run rate. Before sampling, the endpoints are random. After observing data, the computed interval is fixed; the 95% refers to the reliability of the procedure across repetitions, not a 95% probability attached to the fixed parameter.

28.1 Pivot idea

A pivot is a function of data and parameter whose distribution does not depend on unknown quantities. For normal data with known σ ,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

Since

$$\mathbb{P}(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha,$$

rearranging gives

$$\mathbb{P}\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

This probability is over the random sample before data are observed.

28.2 Unknown variance: t interval

For normal data,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1},$$

so a $(1 - \alpha)$ interval is

$$\bar{x} \pm t_{n-1, 1-\alpha/2} \frac{s}{\sqrt{n}}.$$

28.3 Large-sample intervals for non-normal means

By the CLT,

$$\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right),$$

so a large-sample interval uses

$$\bar{x} \pm z_{1-\alpha/2} \frac{s}{\sqrt{n}}.$$

The quality of this approximation depends on sample size and tail behavior.

28.4 Proportion interval

For a sample proportion $\hat{p}$, a basic large-sample standard error is

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

The simple Wald interval $\hat{p} \pm z^* SE$ can perform badly near 0 or 1 or with small samples; score/Wilson or exact-style alternatives are often preferable in practice.

28.5 Three views of confidence intervals

A useful conceptual triad is:

1. **Repeated sampling:** the procedure has coverage $1 - \alpha$.
2. **Test inversion:** the interval contains parameter values not rejected by corresponding level- α two-sided tests.
3. **Precision summary:** estimate $\pm$ uncertainty scale, useful when assumptions support the approximation.

Worked example

A sample of $n = 25$ observations has mean 72 and standard deviation 10. A 95% t interval for the population mean uses 24 df:

$$72 \pm t_{24,0.975} \frac{10}{5} \approx 72 \pm 2.064(2) = [67.87, 76.13].$$

The interval is wider than a known- σ z interval because scale uncertainty is estimated from the same sample.

Exercises

1. Derive the known- σ normal confidence interval from the pivotal inequality.
2. What happens to interval width if n is quadrupled, all else equal?
3. Why is “there is 95% probability that μ is in this computed frequentist interval” generally not the frequentist interpretation?
4. What parameter values are connected to a two-sided 5% hypothesis test through test inversion?

Selected solutions

1. Start from $\mathbb{P}(-z^* \leq (\bar{X} - \mu)/(\sigma/\sqrt{n}) \leq z^*) = 0.95$ and solve the inequalities for μ .
2. Standard error halves.
3. Once data are observed, the fixed-parameter frequentist model treats both parameter and endpoints as fixed; the 95% describes the pre-data procedure.
4. Exactly the values not rejected by the matching two-sided level-0.05 test.

Bootstrap Confidence Intervals

Intuition

The bootstrap asks a daring question: if the unknown population is unavailable, can the observed sample stand in for it? By repeatedly resampling from the data, we imitate repeated sampling and estimate how a statistic varies. This is especially useful when analytic standard-error formulas are difficult or when the statistic is something like a median or percentile.

29.1 Empirical distribution

Given data $x_1, \dots, x_n$, the empirical distribution $\hat{F}_n$ places probability $1/n$ on each observation. A nonparametric bootstrap sample

$$x_1^*, \dots, x_n^*$$

is drawn iid with replacement from $\hat{F}_n$.

29.2 The bootstrap principle

Let $\hat{\theta} = T(X_1, \dots, X_n)$ estimate θ . For each bootstrap sample compute $\hat{\theta}^*$. The bootstrap uses the conditional distribution of

$$\hat{\theta}^* - \hat{\theta}$$

given the observed data to approximate the unknown sampling distribution of

$$\hat{\theta} - \theta.$$

This works when the empirical distribution is a sufficiently good stand-in for the population for the statistic of interest.

29.3 Bootstrap standard error

With B bootstrap estimates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$,

$$\widehat{SE}_{\text{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_b^* - \bar{\hat{\theta}}^*)^2}.$$

29.4 Percentile and basic intervals

If $q_{\alpha/2}^*$ and $q_{1-\alpha/2}^*$ are quantiles of the bootstrap estimates, the percentile interval is

$$[q_{\alpha/2}^*, q_{1-\alpha/2}^*].$$

For the basic bootstrap, compute quantiles d_L, d_U of bootstrap differences $\delta^* = \hat{\theta}^* - \hat{\theta}$ and use

$$[\hat{\theta} - d_U, \hat{\theta} - d_L].$$

The basic interval mirrors the estimated error distribution around the observed estimate.

29.5 Parametric bootstrap

If we assume a parametric model F_θ , estimate $\hat{\theta}$, then simulate bootstrap samples from $F_{\hat{\theta}}$ rather than from the empirical distribution. This can be more efficient if the model is accurate, but inherits its misspecification risk.

Worked example

Suppose we want a confidence interval for the median delivery time from a sample of 80 deliveries. An analytic standard error may be inconvenient. We repeatedly resample 80 observed delivery times with replacement, compute the median of each resample, and use the distribution of those bootstrap medians to quantify uncertainty. The bootstrap is doing experimental repetition on a computer.

Common pitfall

Bootstrap is not magic. It can fail with tiny samples, extreme-tail statistics, strongly dependent data when resampling ignores dependence, parameters on boundaries, or non-smooth estimators. The resampling scheme must respect the data-generating structure.

Listing 29.1: Empirical percentile bootstrap in R

```
B <- 5000
boot_stat <- replicate(B, {
  x_star <- sample(x, size=length(x), replace=TRUE)
  median(x_star)
})
quantile(boot_stat, c(0.025, 0.975))
```

Exercises

1. Why must a standard nonparametric bootstrap sample have the same size n as the original sample?
2. Explain the difference between empirical and parametric bootstrap.
3. If bootstrap estimates are strongly right-skewed, why might a symmetric estimate $\pm 1.96SE$ interval be unattractive?
4. How would you adapt resampling for time-series data where nearby observations are dependent?

Selected solutions

1. It imitates the original sampling experiment: drawing n observations from an estimated population.
2. Empirical bootstrap resamples observed points; parametric bootstrap simulates from a fitted probability model.
3. A symmetric interval ignores asymmetry in the estimated sampling distribution; percentile/basic methods can reflect skew.
4. Use a dependence-aware method such as a block bootstrap instead of resampling single observations.

Simple Linear Regression

Intuition

Regression separates a systematic trend from residual noise. In simple linear regression, the model says that when x changes by one unit, the expected value of Y changes by a constant slope. Least squares chooses the line whose vertical errors have the smallest total squared magnitude.

30.1 Model

The classical simple regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

with the x_i treated as fixed or conditioned on. Then

$$\mathbb{E}[Y_i | x_i] = \beta_0 + \beta_1 x_i.$$

30.2 Least-squares estimates

Choose b_0, b_1 to minimize

$$S(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

The solution is

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Derivation of least squares

Differentiate S :

$$\frac{\partial S}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i) = 0,$$

$$\frac{\partial S}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i) = 0.$$

The first normal equation gives $b_0 = \bar{y} - b_1 \bar{x}$. Substitute this into the second and simplify to obtain the slope formula above. The Hessian is positive semidefinite and positive definite when not all x_i are equal, so this stationary point is the minimum.

30.3 Slope and correlation

Let s_x, s_y be sample standard deviations and r the sample correlation. Then

$$\hat{\beta}_1 = r \frac{s_y}{s_x}.$$

This connects standardized linear association to the unstandardized slope.

30.4 Residuals and R^2

Residuals are $e_i = y_i - \hat{y}_i$. Define

$$SSE = \sum_i e_i^2, \quad SST = \sum_i (y_i - \bar{y})^2.$$

Then

$$R^2 = 1 - \frac{SSE}{SST}.$$

In simple regression with an intercept, $R^2 = r^2$. R^2 describes in-sample variation explained by the fitted linear model; it does not prove causality or predictive validity out of sample.

Worked example

Suppose each extra GB of memory is associated with an estimated 1.8 second reduction in processing time in observed systems. The slope is -1.8 seconds/GB. That is a conditional linear association. Claiming that adding memory *causes* exactly 1.8 seconds of improvement requires a design or assumptions supporting causal interpretation.

30.5 Inference for slope

Under the classical normal-error model,

$$T = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \sim t_{n-2}.$$

Testing $\beta_1 = 0$ asks whether a zero population linear slope is compatible with the data under model assumptions. A confidence interval is

$$\hat{\beta}_1 \pm t_{n-2}^* SE(\hat{\beta}_1).$$

Exercises

1. What does the regression slope measure, including units?
2. Prove that the fitted least-squares line passes through $(\bar{x}, \bar{y})$.
3. If $r = 0.6$, $s_y = 10$, and $s_x = 4$, find the slope.
4. Why does high R^2 not establish causation?

Selected solutions

1. Expected change in response units per one predictor unit under the linear model.
2. The intercept formula gives $\hat{y}(\bar{x}) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$.
3. $0.6(10/4) = 1.5$.
4. Confounding, reverse causation, selection, and purely predictive association can all produce high R^2 .

Multiple Regression, Least Squares Geometry, and Diagnostics

Intuition

Multiple regression lets several predictors explain the response simultaneously. Each coefficient is a partial effect: the change in fitted response associated with one predictor while holding the other included predictors fixed. Geometrically, least squares projects the response vector onto the space spanned by the predictor columns.

31.1 Matrix model

Write

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I).$$

Here X is the design matrix. Least squares minimizes

$$\|\mathbf{y} - X\mathbf{b}\|^2.$$

Differentiating gives the normal equations

$$X^T X \hat{\boldsymbol{\beta}} = X^T \mathbf{y}.$$

If $X^T X$ is invertible,

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}.$$

Orthogonality principle

At the least-squares solution, the residual vector $\mathbf{e} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$ satisfies

$$X^T \mathbf{e} = \mathbf{0}.$$

Thus residuals are orthogonal to every predictor column. This is exactly the geometric condition for $X\hat{\boldsymbol{\beta}}$ to be the orthogonal projection of $\mathbf{y}$ onto the column space of X .

31.2 Interpreting coefficients

In

$$\mathbb{E}[Y \mid X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2,$$

β_1 is the change in expected Y for a one-unit increase in X_1 at fixed X_2 . This “holding fixed” interpretation is central and can differ substantially from the simple-regression slope if predictors are correlated.

31.3 Diagnostics

Important checks include:

- residuals versus fitted values for nonlinearity or changing variance;
- normal quantile plots when small-sample normal-error inference matters;
- influential observations and leverage;
- multicollinearity among predictors;
- out-of-sample validation for prediction.

31.4 Extrapolation and model scope

Regression is most trustworthy within the region of predictor values represented by the data. Extrapolation can fail because the linear relationship may not continue. A mathematically precise fitted equation is still a model, not a physical law.

Worked example

A salary model includes experience and education. The experience coefficient describes salary differences associated with one more year of experience among people with the same modeled education level. If education and experience are correlated, this conditional coefficient can differ from the slope in a salary-versus-experience plot that ignores education.

Exercises

1. Derive the normal equations by differentiating $\|y - Xb\|^2$.
2. What does $X^T e = 0$ mean geometrically?
3. Give one reason a coefficient can change dramatically after adding another predictor.
4. Distinguish an interval for the mean response at x_0 from a prediction interval for a new individual response at x_0 .

Selected solutions

1. Expand $(y - Xb)^T(y - Xb)$; gradient is $-2X^T y + 2X^T Xb$, set to zero.
2. Residuals are orthogonal to the predictor space.
3. The new predictor may be correlated with both the existing predictor and response, altering the conditional relationship.
4. A prediction interval includes both uncertainty in the mean function and new observation noise, so it is wider.

Simulation, Visualization, and R Workflows

Intuition

Computation turns probability from a static formula sheet into an experimental science. Simulation lets you ask “what happens if I repeat this model 100,000 times?” Visualization checks whether assumptions are plausible. R functions encode distributions consistently: `d*` for density/mass, `p*` for CDF, `q*` for quantiles, and `r*` for random generation.

32.1 The d/p/q/r pattern

For the normal distribution:

- `dnorm(x, mean, sd)` gives density;
- `pnorm(x, mean, sd)` gives $\mathbb{P}(X \leq x)$;
- `qnorm(p, mean, sd)` gives the p -quantile;
- `rnorm(n, mean, sd)` generates random observations.

The same pattern holds for `binom`, `pois`, `exp`, `beta`, and others.

Listing 32.1: Simulating a binomial probability

```
set.seed(18)
B <- 100000
x <- rbinom(B, size=20, prob=0.3)
mean(x >= 8) # simulation estimate
pbinom(7, 20, 0.3, lower.tail=FALSE) # exact
```

32.2 Histograms and scatter plots

```
hist(x, probability=TRUE, breaks=20)
plot(df$x, df$y)
fit <- lm(y ~ x, data=df)
abline(fit)
plot(fit) # diagnostic plots
```

A histogram answers questions about shape and scale. A scatter plot answers questions about relationship. Always plot before trusting a model whose assumptions are visible in the plot.

32.3 Bayesian grid update

For a discrete grid of possible p values:

```
p <- seq(0.01, 0.99, length.out=99)
prior <- rep(1/length(p), length(p))
lik <- dbinom(7, size=10, prob=p)
post <- prior * lik
post <- post / sum(post)
```

This is continuous Bayes approximated on a fine grid and is excellent for intuition.

32.4 Classical tests and intervals

```
t.test(x, mu=0)
t.test(x, y) # Welch two-sample default
chisq.test(table(group, outcome))
fit <- lm(y ~ x1 + x2, data=df)
summary(fit)
confint(fit)
```

Software returns numbers quickly; your job is to know what null hypothesis, model, standard error, and interpretation produced those numbers.

32.5 Monte Carlo error

If a simulated event has probability p and we use B independent simulation runs, the Monte Carlo proportion has standard error

$$\sqrt{\frac{p(1-p)}{B}}.$$

Simulation itself has sampling uncertainty. Increasing B by a factor of 100 reduces Monte Carlo standard error by a factor of 10.

Exercises

1. Which R function computes the 97.5th percentile of a standard normal distribution?
2. Write pseudocode for estimating $\mathbb{P}(X + Y > 3)$ for independent exponential variables by simulation.
3. Why should `set.seed()` be used in reproducible analysis?
4. If a Monte Carlo estimate of a probability near 0.5 uses $B = 10,000$, approximately what is its standard error?

Selected solutions

1. `qnorm(0.975)`.
2. Repeatedly draw X, Y , record the indicator of $X + Y > 3$, and average indicators.
3. It makes the pseudo-random sequence reproducible.
4. Approximately $\sqrt{0.25/10000} = 0.005$.

Part V

Deep Connections and Exam-Level Synthesis

A Unifying View: Four Distributions You Must Not Confuse

Intuition

Many statistical mistakes come from confusing distributions that live at different levels. Keep four objects separate: the population/data distribution, the likelihood, the sampling distribution of a statistic, and the posterior distribution of a parameter. They may use the same algebraic ingredients but answer different questions.

33.1 Population distribution

$X \sim f(x \mid \theta)$ describes variability in observations given a fixed parameter. Example: individual heights around a population mean.

33.2 Likelihood

$L(\theta; x) = f(x \mid \theta)$ treats observed x as fixed and compares parameter values. It need not normalize over θ .

33.3 Sampling distribution

$\hat{\theta} = T(X_1, \dots, X_n)$ varies across hypothetical repeated datasets at fixed θ . Its distribution determines bias, standard errors, p-values, and confidence intervals.

33.4 Posterior distribution

$\pi(\theta \mid x)$ describes Bayesian uncertainty about the parameter after observing x and combining likelihood with a prior.

Checkpoint

When reading any formula, ask: **what is random, what is fixed, and what is being conditioned on?** If you can answer those three questions, most probability/statistics notation becomes much easier.

Modeling Checklist: From Story to Mathematics

A reliable workflow for almost any 18.05 problem is:

1. **Name the experiment.** What is observed? What can vary?
2. **Define random variables.** Include units.
3. **State assumptions.** Independence? identical distribution? normality? fixed n ? rare events?
4. **Choose the distributional story.** Bernoulli, binomial, Poisson, normal, etc.
5. **Identify the target.** Probability, expectation, parameter, interval, test, prediction?
6. **Choose the mathematical tool.** Counting, conditioning, Bayes, expectation, CLT, likelihood, bootstrap, regression.
7. **Compute with units and scale checks.**
8. **Interpret in the original story.**
9. **Stress-test assumptions.** What changes if independence, normality, or the prior is wrong?

Worked example

Story: “In 1,000 visits, 230 users convert. Is the true conversion rate above 20%?”

Experiment: 1,000 Bernoulli outcomes. Model: iid Bernoulli(p) as a first approximation. Target: evidence about $p > 0.2$. Frequentist route: one-sided proportion test or large-sample z-statistic. Bayesian route: choose a prior for p , update to a beta posterior, and compute $\mathbb{P}(p > 0.2 \mid x)$. Both analyses require asking whether visits are independent and representative.

Comprehensive Mixed Practice Set

The problems below deliberately mix topics. Try to identify the correct probabilistic object before calculating.

Problems 1–12

1. A password has 10 characters, each independently chosen from 26 lowercase letters. What is the probability it contains no repeated character?
2. A disease has prevalence 0.002. A test has 99% sensitivity and 1% false-positive rate. Find the posterior disease probability after a positive result.
3. Let $X \sim \text{Pois}(4)$. Compute $\mathbb{P}(X \geq 2)$ and give a real-world interpretation.
4. If X_i are iid with mean 5 and variance 16, approximate $\mathbb{P}(4.8 < \bar{X}_{400} < 5.2)$ using the CLT.
5. A prior for a Bernoulli success rate is $\text{Beta}(4, 6)$. After 30 trials with 21 successes, find the posterior distribution and posterior mean.
6. Data $X_i \sim \mathcal{N}(\mu, 9)$ with known variance. For $n = 36$ and $\bar{x} = 11$, construct a 95% confidence interval for μ .
7. In a two-sided z-test, $z = 3.0$. Give the p-value expression and explain what it means without using the phrase “probability the null is true.”
8. A sample correlation is $r = -0.8$. What does this say and what does it not say?
9. Explain how a bootstrap confidence interval for the 90th percentile of a latency distribution would be constructed.
10. Derive the MLE for the mean of a normal distribution with known variance.
11. In a regression $\hat{y} = 12 + 3.5x$, interpret 3.5. Name one reason it may not be causal.
12. A one-way ANOVA rejects equal means for five groups. What can and cannot be concluded immediately?

Solutions 1–12

1. Favorable strings have $26 \cdot 25 \cdots 17$ choices; total 26^{10} , so probability is $(26)_{10}/26^{10}$.
2. $0.99(0.002)/[0.99(0.002) + 0.01(0.998)] \approx 0.1656$.
3. $1 - \mathbb{P}(0) - \mathbb{P}(1) = 1 - e^{-4}(1 + 4)$. It is the chance of at least two events in a window with mean count 4 under the Poisson assumptions.
4. $SE = 4/20 = 0.2$. Standardized bounds are -1 and 1, so probability is approximately $\Phi(1) - \Phi(-1) \approx 0.6827$.
5. $\text{Beta}(25, 15)$; mean $25/40 = 0.625$.
6. $11 \pm 1.96(3/6) = 11 \pm 0.98 = [10.02, 11.98]$.
7. $2[1 - \Phi(3)] \approx 0.0027$. Under the null model, a statistic at least this extreme in absolute value would occur about 0.27% of repeated samples.

8. Strong negative linear association in the sample; it does not prove causality and does not summarize nonlinear association fully.
9. Resample the observed latencies with replacement many times, compute the 90th percentile each time, and form a percentile/basic interval from the bootstrap distribution.
10. The log-likelihood is constant minus $\sum_i (x_i - \mu)^2 / (2\sigma^2)$; minimizing squared deviations gives $\hat{\mu} = \bar{x}$.
11. Each additional unit of x is associated with 3.5 additional response units in the fitted linear model. Confounding is one noncausal explanation.
12. There is evidence that not all five population means are equal under ANOVA assumptions. The test does not identify which means differ or establish practical importance.

Problems 13–24

13. Prove $\text{Cov}(X, X) = \text{Var}(X)$ and deduce $\text{Corr}(X, X) = 1$ when variance is positive.
14. Let X, Y be independent exponentials with rates 2 and 5. Find $\mathbb{P}(X < Y)$.
15. A fair coin is flipped until the first head. Find the expected number of flips and variance.
16. If $X \sim \text{Unif}(0, 2)$ and $Y = X^3$, derive the CDF of Y .
17. For iid X_i with finite variance, prove $\bar{X}_n$ is consistent using Chebyshev's inequality.
18. Explain why optional stopping or repeated peeking at p-values can inflate false-positive rates under naive testing.
19. A 95% CI for a mean is $[10, 14]$. Is $H_0 : \mu = 15$ rejected by the corresponding two-sided 5% test? Explain.
20. A $\text{Beta}(1, 1)$ prior and 0 successes in 5 trials produce what posterior? Why is the posterior mean nonzero?
21. What feature of the exponential distribution makes it a natural waiting-time model for a Poisson process?
22. Explain why regression residual plots can reveal model problems that R^2 hides.
23. Compare the uncertainty represented by a confidence interval for the mean response and a prediction interval for a new response.
24. Design a simulation to verify the central limit theorem for a highly skewed exponential population.

Solutions 13–24

13. $\text{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \text{Var}(X)$; dividing by σ_X^2 gives correlation 1.
14. $\mathbb{P}(X < Y) = \int_0^\infty 2e^{-2x} \mathbb{P}(Y > x) dx = \int_0^\infty 2e^{-7x} dx = 2/7$.
15. Geometric($p = 1/2$) on $1, 2, \dots$: mean 2, variance $(1 - p)/p^2 = 2$.
16. For $0 \leq y \leq 8$, $F_Y(y) = \mathbb{P}(X^3 \leq y) = \mathbb{P}(X \leq y^{1/3}) = y^{1/3}/2$.
17. $\text{Var}(\bar{X}_n) = \sigma^2/n$ and Chebyshev gives $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$.
18. Repeated looks create many chances to cross a nominal rejection threshold; the overall stopping rule changes the null distribution of the decision procedure.
19. Yes. By test-interval duality, parameter values outside the matching 95% confidence interval are rejected at two-sided level 5%.
20. $\text{Beta}(1, 6)$; posterior mean $1/7$. The prior and finite data do not imply certainty that $p = 0$.
21. Memorylessness: after waiting without an event, the remaining waiting-time distribution is unchanged.
22. Residuals can show curvature, heteroskedasticity, clusters, and influential points even when overall variance explained is high.

23. Mean-response intervals quantify uncertainty about the conditional mean; prediction intervals additionally include irreducible individual noise and are wider.
24. For each n such as 1,5,20,100, repeatedly draw n exponential observations, compute standardized sample means, and compare histograms/quantiles with $N(0, 1)$.

Part VI

Reference Appendices

Proof Toolbox

A.1 Linearity of expectation

For constants a_i and integrable random variables X_i ,

$$\mathbb{E} \left[\sum_i a_i X_i \right] = \sum_i a_i \mathbb{E}[X_i].$$

No independence is required.

A.2 Markov and Chebyshev inequalities

For nonnegative X and $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}.$$

Proof: $X \geq a \mathbf{1}\{X \geq a\}$, then take expectations.

Applying Markov to $(X - \mu)^2$ gives Chebyshev:

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

These inequalities require little distributional knowledge and are the engine behind an elementary proof of the weak LLN.

A.3 Jensen's inequality

If φ is convex,

$$\varphi(\mathbb{E}X) \leq \mathbb{E}[\varphi(X)].$$

Geometrically, a convex graph lies below its chords, so averaging inputs before applying a convex function cannot exceed averaging function values. Variance nonnegativity can be viewed through convexity of x^2 .

A.4 Cauchy-Schwarz

For square-integrable variables,

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

Applying this to centered variables bounds correlation by 1.

A.5 Law of total expectation and variance

$$\mathbb{E}X = \mathbb{E}[\mathbb{E}(X \mid Y)],$$

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

The second formula decomposes uncertainty into average within-group uncertainty plus between-group uncertainty.

Law of total variance

Let $m(Y) = \mathbb{E}[X \mid Y]$. Write

$$X - \mathbb{E}X = (X - m(Y)) + (m(Y) - \mathbb{E}X).$$

Square and take expectations. The cross term vanishes because

$$\mathbb{E}[X - m(Y) \mid Y] = 0.$$

The two remaining terms become $\mathbb{E}[\text{Var}(X \mid Y)]$ and $\text{Var}(m(Y))$.

Distribution Atlas

Distribution	PMF/PDF	Mean	Variance	Use story
Bernoulli(p)	$p^x(1-p)^{1-x}$, $x \in \{0, 1\}$	p	$p(1-p)$	one binary trial
Binomial(n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	successes in fixed n
Geometric(p)	$(1-p)^{k-1}p$	$1/p$	$(1-p)/p^2$	wait for first success
Poisson(λ)	$e^{-\lambda}\lambda^k/k!$	λ	λ	event count in exposure
Uniform(a, b)	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$	equal density on interval
Exponential(λ)	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	memoryless waiting time
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2	additive variation / CLT
Beta(a, b)	$p^{a-1}(1-p)^{b-1}/B(a, b)$	$a/(a+b)$	$ab/[(a+b)^2(a+b+1)]$	prior/posterior for probability
χ^2_ν	sum of ν squared standard normals	ν	2ν	variance/count discrepancy
t_ν	normal divided by estimated scale	0 for $\nu > 1$	$\nu/(\nu-2)$ for $\nu > 2$	mean inference, unknown SD
F_{d_1, d_2}	ratio of scaled chi-squares	$d_2/(d_2-2)$ if $d_2 > 2$	varies	variance ratio, ANOVA

Formula Atlas

Counting

$$P(n, k) = \frac{n!}{(n-k)!}, \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Probability

$$\begin{aligned} P(A^c) &= 1 - P(A), \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B), \\ P(A | B) &= \frac{P(A \cap B)}{P(B)}, \\ P(H_i | E) &= \frac{P(E | H_i)P(H_i)}{\sum_j P(E | H_j)P(H_j)}. \end{aligned}$$

Moments

$$\begin{aligned} E[g(X)] &= \sum_x g(x)p(x) \quad \text{or} \quad \int g(x)f(x)dx. \\ \text{Var}(X) &= E[X^2] - (EX)^2. \\ \text{Cov}(X, Y) &= E[XY] - EX EY. \\ \text{Var}(X + Y) &= \text{Var } X + \text{Var } Y + 2 \text{Cov}(X, Y). \end{aligned}$$

Sample mean

$$\begin{aligned} E\bar{X} &= \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}. \\ \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} &\approx N(0, 1). \end{aligned}$$

MLE

$$L(\theta) = \prod_i f(x_i | \theta), \quad \ell(\theta) = \sum_i \log f(x_i | \theta).$$

Bayes

$$\begin{aligned} \pi(\theta | x) &\propto L(\theta; x)\pi(\theta). \\ \text{Beta}(a, b) + s \text{ succ.}, f \text{ fail.} &\Rightarrow \text{Beta}(a + s, b + f). \end{aligned}$$

Tests

$$\begin{aligned} Z &= \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \quad T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}. \\ X^2 &= \sum \frac{(O - E)^2}{E}. \\ F &= \frac{MS_B}{MS_W}. \end{aligned}$$

Intervals

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}, \quad \bar{x} \pm t^* \frac{s}{\sqrt{n}}.$$

Regression

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}. \\ \hat{\beta} &= (X^T X)^{-1} X^T y. \end{aligned}$$

Conceptual Traps and How to Avoid Them

1. **Density is not probability.** Integrate density over an interval to obtain probability.
2. **Independence is not disjointness.** Nonempty disjoint events with positive probability cannot be independent.
3. **Zero correlation is not independence.** Correlation detects linear association only.
4. **Expected value need not be a possible outcome.** A fair die has expectation 3.5.
5. **LLN is not CLT.** LLN gives concentration; CLT gives the shape and scale of fluctuations.
6. **Likelihood is not a probability distribution over parameters.** A prior is needed to turn it into a Bayesian posterior.
7. **A p-value is not $P(H_0 \mid x)$.** It is a tail probability computed under H_0 .
8. **Non-significant is not proof of no effect.** Power may be low.
9. **Statistical significance is not practical importance.** Report effect magnitude and uncertainty.
10. **A confidence interval is not automatically a credible interval.** Their probability statements condition on different objects.
11. **Bootstrap must preserve structure.** IID resampling is wrong for clustered or time-dependent observations unless adapted.
12. **Correlation and regression do not create causality.** Causal claims require design or additional assumptions.
13. **High R^2 does not prove a good model.** Residual patterns and out-of-sample performance matter.
14. **Model formulas are conditional on assumptions.** Independence, sampling design, and distributional shape are part of the mathematics, not optional prose.

Study Roadmap

Pass 1: build intuition

Read Chapters 1–13 quickly, focusing on stories: counting, conditioning, random variables, expectation, distributions, LLN/CLT, and dependence. You should be able to describe each concept without formulas.

Pass 2: learn inference as two reverse problems

Read Chapters 14–24. Keep asking what is fixed and what is random. Derive at least one MLE, one beta posterior, one z/t statistic, and one chi-square statistic by hand.

Pass 3: uncertainty and modeling

Read confidence intervals, bootstrap, and regression. Practice translating a word problem into a model before computing anything.

Pass 4: exam preparation

Use the mixed practice, distribution atlas, formula atlas, and conceptual-traps chapter. Redo every exercise you missed without looking at the solution. For each formula, state the assumptions aloud.

Course Coverage Reference

Topic coverage was checked against the official MIT OpenCourseWare materials for *18.05 Introduction to Probability and Statistics*, including the syllabus, class calendar, and class reading/in-class-material map. This guide is independently written and is not an official MIT publication.

End of guide.